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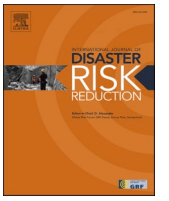

# BIM and GIS-integrated multi-stakeholder fire risk assessment model for heritage buildings

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ABSTRACT

Heritage buildings are vulnerable to fires due to combustible materials, outdated suppression systems, and conservation constraints. Systematic fire risk assessment is therefore essential to appraise building conditions, identify dangers early, and enable proactive mitigation. However, current practice suffers from insufficient risk identification, limited multi-stakeholder involvement, and low integration with digital technologies. This study aims to develop RATER, a multi-stakeholder fire risk assessment model integrating AHP-derived expert weights with data-driven analysis for heritage buildings. RATER identifies 17 fire risk factors across fire hazards, heritage and firefighting equipment, surrounding environment, and fire safety management categories. To quantify risk weights, a pairwise comparison survey involving 53 professionals from fire services, heritage management, conservation, and safety engineering domains is conducted. The analytic hierarchy process (AHP) is applied to derive weighted priorities of fire risk factors and reveal consensus and divergence in expert judgments. The study further conducts professional-centric assessment and data-driven analysis using building information modeling (BIM) and geographic information systems (GIS) to display quantitative geo-spatial risk distribution. Through this dual assessment, the model is validated on three UNESCO heritage buildings in Macao. The results indicate that ignition sources, combustible materials, and integrated fire service installations are the top-ranked risk factors. Moreover, professional-centric and BIM-GIS data-driven results present high consistency in risk quantification, which reinforces the validity of the findings and offers complementary insights. RATER provides a structured fire risk model supported by multi-stakeholder insights and digital technology integration. Practically, it enables managers and policymakers to identify high-risk areas and support proactive mitigation planning.

## 1. Introduction

Heritage buildings possess significant value for preserving architectural integrity and historical authenticity. However, these

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buildings are particularly vulnerable to fire hazards. Unlike modern buildings mandated to incorporate non-combustible materials, fire-resistant compartmentation, and integrated suppression systems (e.g., sprinklers and smoke detectors), heritage buildings often contain timber frameworks, flammable furnishings, and open spatial configurations that accelerate fire spread. Moreover, conservation regulations tend to prohibit intrusive retrofitting of active fire protection systems. This leaves heritage buildings reliant on passive measures that may be compromised by material deterioration over time.

Recent incidents have demonstrated the severe vulnerability of cultural heritages to fire hazards. For instance, in May 2024, the illegal use of a gas torch during renovation work at Henan University's Auditorium ignited its wooden roof [1]. The resulting fire caused a partial collapse of this nationally protected heritage site. The incident exposed several serious deficiencies and failures, including excessive storage of flammable materials, expired firefighting equipment, and inadequate evacuation routes within the structure. Similarly, in March 2025, a major fire in Lima's Barrios Altos Historic District destroyed four colonial-era buildings [2]. Despite the efforts of more than 400 firefighters, containment was hindered by limited hydrant access, narrow streets blocking fire trucks, and the rapid spread of flames due to inadequate spacing between buildings. These incidents highlight the urgent need for comprehensive fire risk assessments to identify existing weaknesses and develop proactive protection measures that safeguard irreplaceable heritage assets for future generations.

Current practice involves fire safety or heritage professionals conducting on-site inspections. Given the multifaceted nature of fire risks in heritage buildings, structured frameworks are instrumental in guiding the inspection procedures. Specifically, several established frameworks, such as fire risk assessment method for engineering (FRAME) [3] and fire risk assessment for old city centers (ARICA) [4], are widely utilized for risk level and heritage vulnerability evaluation. These frameworks have distinct structures, methods, and criteria. For example, FRAME applies engineering-based quantitative risk calculations focusing on technical parameters such as fire load density and building characteristics [5–7], while ARICA adopts a regulatory compliance approach to structure fire risks into sequential fire stages (ignition, propagation, evacuation, and combat) for urban-scale assessments [8,9]. These frameworks have been applied in real-world heritage buildings to support risk identification and fire safety planning. Liao et al. [3] utilized an improved FRAME method to assess an ancient Chinese village in terms of building proximity, traditional customs, fire protection layout, electrical installations, and structural characteristics. Similarly, Granda and Ferreira [4] applied the ARICA method to evaluate a historic center in Portugal across the dimensions of fire ignition, propagation, evacuation capabilities, and combat effectiveness. Overall, these frameworks enable the identification of key risk factors and provide valuable guidance for structured fire risk assessments.

Apart from risk assessment frameworks, emerging technologies such as building information modeling (BIM) and geographic information systems (GIS) have been increasingly adopted for fire risk modeling and spatial analysis. BIM provides detailed 3D representations of building structures and embedded fire safety systems, and GIS enables data access to surrounding environments, road networks, and external firefighting resources. In existing studies, BIM has been utilized for simulating fire scenarios and optimizing evacuation routes [10–15]. Meanwhile, GIS is widely used for city-level investigations, focusing on spatial distribution of fire risks and resource allocation [4,16–18]. Moreover, the integration of BIM and GIS is gradually being explored to combine building- and urban-scale data for fire risk visualization [10,19,20]. These prior studies have demonstrated the potential of BIM and GIS to enable data-driven, multi-scale fire risk analysis across diverse built environments.

Despite these advances, three primary challenges remain, i.e., (1) incomplete risk factor coverage, (2) insufficient multi-stakeholder involvement, and (3) limited digital technology integration. First, existing assessments often fail to comprehensively address all relevant fire risk factors. External environmental threats such as access restrictions and natural hazards, as well as safety management factors like maintenance routines have been relatively underestimated in prior research [21]. This restricts their ability to provide holistic and actionable insights. Second, current assessments often lack interdisciplinary input. They are usually conducted by one or two professional groups, such as fire engineers or academic researchers, which may lead to biased evaluations due to limited viewpoints. Third, regarding the adoption of BIM and GIS, although existing studies have increasingly leveraged them in fire risk research, their integration into a unified risk assessment framework for data-driven analysis has not been sufficiently explored. Substantial efforts are still needed in data mapping, standardized fire risk calculation, and automation for real-world applications. Specifically, mapping between the requirements of fire risk modeling and the available information in BIM and GIS is essential to extract necessary data while filtering out redundant details. Standardized methods are also needed to formulate quantitative risk indicators and convert BIM and GIS data into these indicators. Automating these processes is particularly important, yet still lacking, to ensure the practical adoption in fire risk assessment. Together, these three challenges result in none of the existing solutions being able to fully integrate multi-stakeholder inputs with BIM-GIS data.

Given these challenges, this study develops RATER, a structured fire risk assessment model for heritage buildings that integrates multi-stakeholder insights with BIM-GIS data analysis. It incorporates 17 fire risk factors from fire hazards, heritage and firefighting equipment, surrounding environment, and fire safety management categories. Four professional groups comprising fire emergency services, heritage building management, cultural heritage conservation, and building safety engineering are involved in conducting factor weighting analysis using the analytic hierarchy process (AHP) method. In addition, the study conducts both professional-centric and data-driven assessments by integrating BIM and GIS technologies to quantify fire risks and develop a 3D system for visualizing geospatial risk distribution. The proposed RATER and the BIM-GIS system are validated through cross-comparison between professional-centric and data-driven assessments conducted on three heritage buildings in the historic center of Macao, a UNESCO World Heritage Site. The professional-centric assessment collects risk scores directly from practitioners through structured surveys, while the data-driven assessment quantifies risk scores objectively using spatial and geometric attributes extracted from BIM models and GIS databases. Since the two assessments originate from independent data sources (i.e., professional judgments and spatial data calculations), the dual assessments are adopted to cross-validate the feasibility of RATER and reinforce the findings.

**Table 1**
Fire safety assessment factors in existing models.

| Category | Factor | ARICA | FRAME | FRIM-HB | CHICHORRO | PFDI |
|---|---|---|---|---|---|---|
| Fire hazards | Ignition sources | ● | ● | ● | ● | ● |
| | Combustible and flammable materials | ● | ● | ● | ● | ● |
| | Natural hazards | ● | ● | ● | ● | ● |
| Heritage and firefighting equipment | Building inherent features | ● | ● | ● | ● | ● |
| | Means of egress | ● | ● | ● | ● | ● |
| | Portable firefighting devices | ● | ● | ● | ● | ● |
| | Integrated fire service installations | ● | ● | ● | ● | ● |
| | Water supplies | ● | ● | ● | ● | ● |
| Surrounding environment | External space characteristics | ● | ● | ● | ● | ● |
| | Road traffic | ● | ● | ● | ● | ● |
| | Firefighting resources | ● | ● | ● | ● | ● |
| | Surrounding water supplies | ● | ● | ● | ● | ● |
| | Surrounding potential dangers | ● | ● | ● | ● | ● |
| Fire safety management | Occupancy status | ● | ● | ● | ● | ● |
| | Emergency plans | ● | ● | ● | ● | ● |
| | Routine maintenance | ● | ● | ● | ● | ● |
| | Multi-stakeholder training | ● | ● | ● | ● | ● |

*Note.* Dots in green, yellow, and red represent comprehensive, partial, and no consideration in existing fire risk assessment models, respectively.

The remainder of the paper is structured as follows. Section 2 reviews existing fire risk assessment frameworks, professional involvement, and BIM- and GIS-driven approaches. Section 3 describes the RATER development and associated professional-centric and data-driven methods. Section 4 presents the assessment results of real-world case studies. Section 5 discusses the key findings regarding multi-stakeholder risk weighting, comparison between professional-centric and data-driven assessments, and geo-spatial insights within the integrated system. Section 6 concludes with research implications and future directions.

## 2. Literature review

### 2.1. Fire risk assessment models for heritage buildings

Fire risk assessment for heritage buildings involves multiple dimensions, including fire hazards, building characteristics, fire service installations, surrounding resources and threats, and safety management. In current practice, numerous established frameworks exist for various applications such as risk assessments, compliance checking, and fire engineering analysis [22]. However, not all these frameworks are directly adaptable to fire risk assessments for heritage buildings. Accordingly, several criteria are established to refine the review scope and select the most relevant models: (1) the method should constitute a comprehensive multi-factor assessment framework, (2) it should be sufficiently adaptable to diverse heritage buildings and designed for fire risk assessment, and (3) it should have documented applications to heritage buildings or historic urban areas in recent years. Consequently, five representative models were selected for further comparison and analysis: FRAME [3], ARICA [4], fire risk index method for heritage buildings (FRIM-HB) [23], construction holistic fire risk calculation and optimization of its reduction with construction works (CHICHORRO) [24,25], and the potential fire damage index (PFDI) [26]. These models provide structured frameworks to quantify the fire risk levels from multiple perspectives.

First, FRAME focuses on physical characteristics of buildings and fire prevention facilities. It was initially designed to assess fire risks in industrial and commercial buildings, covering technical parameters such as fire load density, building size, and fire protection facilities. The risk quantification process comprehensively considers the potential risk, the acceptable level, and the protection level. Recent studies have applied FRAME to various historical structures, such as three historic churches in Porto [7] and heritage buildings in Viseu's historic center [6].

Different from FRAME, ARICA was developed for historic urban areas and heritage buildings in accordance with the fire safety regulations in Portugal. It focuses on assessing the condition of building protection, the effectiveness of evacuation systems, and the completeness of firefighting facilities to determine the fire risk level. Fire ignition, fire propagation, evacuation, and fire combat are considered together for comprehensive risk assessment. ARICA facilitates consistent risk evaluation across extensive building stocks, thereby providing quantitative risk indexing for urban-scale spatial analysis. This model has been widely applied in several historical districts across Portugal and Ecuador [4,9,22,27,28].

FRIM-HB employs AHP methods to establish an evaluation framework and determine the weights of various risk factors through a Delphi expert panel consultation process. FRIM-HB includes 23 factors covering building protection, evacuation, firefighting facilities, and maintenance. Specifically, FRIM-HB emphasizes heritage characteristics, such as historical values, building monitoring, and potential sources of danger. It was successfully applied to the Italian cultural heritage risk mapping system (Carta del Rischio [23]), which demonstrates its capabilities in the preventive protection of heritage buildings.

CHICHORRO adopts a holistic approach by integrating four dimensions of fire risk analysis (i.e., the probability of fire occurrence, fire consequence, fire development and propagation, and the efficiency of fire combat and safety) into a unified framework. The probability evaluates how building characteristics and daily operations influence the likelihood of fire occurrence; the consequence considers compartments and evacuation routes; the fire development focuses on building structure and fire separations; and the efficiency assesses fire facilities and emergency management. By synthesizing these dimensions together, it comprehensively encompasses fire initiation, growth and spread, control measures, and overall impacts in line with fire safety principles. In practice, CHICHORRO has been applied to projects in the Berlengas neighborhood of Porto [24] and the urban district of Braga [25] to assist in fire risk mapping and renovation planning.

PFDI was specifically developed for vulnerability assessment of heritage buildings [26]. It emphasizes material degradation, utility conditions, and emergency preparedness, rather than fire occurrence probability. PFDI comprises 21 indicators organized into four domains: building elements, utility infrastructure, fire protection systems, and emergency planning. The assessment adopts a weighted weakest-link principle, highlighting the decisive influence of poorly performing components on overall vulnerability. The core innovation of this method lies in quantifying potential damage rather than occurrence probability. Such an approach offers a complementary perspective to probability-based models. PFDI has been applied to the Misericórdia Church in Esposende, where electrical system renovations and personnel training significantly reduced the fire damage index, demonstrating its diagnostic value in incremental risk mitigation [26].

Table 1 summarizes key factors in the above-mentioned models in terms of fire hazards, heritage and firefighting equipment, surrounding environment, and fire safety management dimensions. All five models place significant emphasis on building and firefighting equipment, such as building characteristics, means of egress, fire service installations, and water supplies. Regarding fire hazards, ignition sources and flammable materials have also been comprehensively considered, except that natural disasters are less covered. Although prior studies such as [26,29] have investigated lightning-induced and earthquake-induced fires, they treated these natural hazards only as implicit triggering stimuli rather than independent factors, which may underestimate their impact on heritage buildings. By contrast, surrounding environment factors received relatively less attention in the past, with more explicit consideration only emerging in recent publications. Several factors such as road traffic and external danger remain partially integrated, leaving room

for developing more unified and comprehensive assessments. Lastly, the safety-management dimension is not sufficiently addressed. Although factors such as emergency planning and routine maintenance appear in existing models, they are typically treated without multi-stakeholder considerations and are embedded within system-level assessments, limiting the explicit representation of human safety management practices. These findings reveal critical blind spots in existing models, indicating a clear need for a more comprehensive heritage fire risk assessment model, with more attention to the surrounding environment and operational management dimensions.

### 2.2. Stakeholder-involved fire risk weighting and assessment

Fire risk assessments involve diverse domain-specific knowledge pertaining to building characteristics, conservation methods, emergency preparedness, and safety regulatory compliance. Given this multifaceted nature, stakeholder involvement is prevalent and crucial to ensure assessments are grounded in practice. The participation of professionals such as fire safety engineers, heritage conservators, building managers, and emergency responders can bring valuable insights based on their distinct expertise and contribute to comprehensive risk identification, weighting, and assessment.

Many existing studies have engaged different professionals in the risk assessment process. For instance, Wang et al. [30] collaborated with ancient building professionals to identify fire risk and assign weight factors via AHP for assessments of the Imperial Palace in Beijing. Revez et al. [31] surveyed twelve conservation professionals and three government supervisors to validate damage risk assessments for masonry heritage buildings in Spain and Portugal. Beyond heritage buildings, Mi et al. [32] engaged six safety engineering professionals to construct triangular fuzzy judgment matrices for ranking factor priorities for 3 residential buildings. Similarly, Ketsakorn and Phangchandha [33] assembled seven practitioners from building management and fire safety domains to carry out fire risk assessments for school buildings through the Delphi method. Bakhtiyari et al. [34] established a committee of 20 firefighting professionals to identify and score parameters within the existing building regulatory framework of Iran. Alao et al. [35] integrated opinions of thirty Nigerian fire safety practitioners and professionals to collectively evaluate ten components in fire safety management for office building risk assessment in Abuja. Overall, existing studies dominantly rely on inputs from fire safety or building management professionals, yet the engagement of heritage conservators remains relatively rare in current studies. Nevertheless, professionals in different domains often hold divergent values and priorities. For example, fire safety engineers may emphasize active suppression systems and regulatory compliance, whereas conservators prioritize the preservation of original materials and architectural authenticity. Meanwhile, building managers and emergency responders may contribute practical insights related to daily operations, maintenance limitations, or emergency access. Without explicitly understanding and balancing these differing viewpoints, fire risk assessments may become incomplete due to the narrowed disciplinary scope, leaving certain vulnerabilities unaddressed.

In recent years, multi-stakeholder assessments have gradually received attention to enable collective thinking from professionals in different disciplines for balancing different considerations. For instance, Wu et al. [36] calculated weights of fire risk factors in Chengkan village by employing the analytic network process method to integrate opinions from eight professionals in natural disasters, heritage village protection, fire rescue, and construction engineering safety. Also, Li et al. [37] engaged a diverse group of seven Chinese professionals, including resident representatives, firefighters, researchers, and government fire management officials, to score indicators through triangular fuzzy evaluation. These examples show the potential advantages of integrating diverse perspectives to achieve more holistic and nuanced risk assessments. Nevertheless, this remains a developing area with several research gaps.

(1) The explorations of multi-stakeholder involvement are still limited in both scope and depth, as only a few studies have attempted such integration and often with a small number of experts.
(2) Among these explorations, most studies tend to consolidate inputs of multiple professionals into a single aggregated result, without systematically investigating the underlying differences in their risk considerations. As a result, the understanding of how professionals differ in their risk perceptions and priorities is still unclear, and the opportunities for their perspectives to complement one another in risk assessments have not been thoroughly explored.

Therefore, a systematic investigation is needed to clarify professional differences and harness their complementary insights for more effective fire risk assessments.

### 2.3. Data-driven risk modeling and visualization methods

In addition to professional-based evaluations, another prevalent approach to fire risk assessment is data-driven analysis. Unlike professional-based methods that rely on professional yet subjective judgments, data-driven approaches utilize objective metrics to quantify risk factors in a systematic and replicable manner. To support such analysis, BIM and GIS technology emerge and provide abundant data to facilitate the risk assessment process [38].

BIM has been widely adopted for fire risk quantification, simulation, and emergency planning in both modern and heritage buildings [11,39]. Wang et al. [40] developed a BIM-enabled method that extracts building material attributes, spatial functions, and fire resistance ratings from industry foundation classes (IFC) files to compute standardized fire risk indices during operation and maintenance phases. Sun and Turkan [41] expanded the application of BIM by integrating it with fire dynamics simulator (FDS) and agent-based modeling to simulate fire spread and occupant evacuation scenarios. Wong and Lee [42] developed an IFC-based approach that enriches BIM models with emergency-specific information to support multi-agency coordination during indoor fire emergencies. These studies demonstrate that BIM can serve as a comprehensive, structured repository to manage various building data and facilitate

quantitative analysis of fire risks in heritage buildings [43].

Parallel to BIM technology, GIS provides powerful capabilities for analyzing spatial fire risk factors related to external environments [12,44–46]. Tikhonova et al. [16] leveraged GIS-based systems to support fire vulnerability mapping in urban heritage districts by collecting spatial features such as road width, building proximity, and terrain characteristics. Lei et al. [18] proposed a hybrid GIS-machine learning approach using entropy-weighted indicators and XGBoost algorithms to predict monthly fire risk levels of timber heritage buildings. Anderson-Bell et al. [17] used high-resolution aerial imagery, digital terrain models, and census data to perform deep learning-based predictions on non-residential building fire risks in urban areas. Granda and Ferreira [4] conducted a GIS-based fire vulnerability assessment of over 400 heritage buildings in Guimarães and mapped risk levels across urban zones. These applications highlight the ability of GIS to support scalable, data-driven modeling of fire risks in complex urban and suburban environments.

The integration of BIM and GIS has shown great potential to provide comprehensive risk modeling by capturing both building internal and external factors. Qiao et al. [19] proposed a framework that combines BIM and GIS to interpret and prioritize fire risks based on both internal vulnerabilities and surrounding hazards. Similarly, Meschini et al. [10] created a combined BIM-GIS platform for university buildings to simulate crowd movement and map escape routes. Sammartano et al. [20] developed an integrated HBIM-GIS vulnerability assessment approach that combines building geometric and semantic information with structural data for multi-scale damage and vulnerability evaluation.

While existing studies have showcased the capabilities of BIM and GIS technologies for data-driven risk assessments, several limitations still exist and hinder their real-world adoption.

(1) There is a low level of integration between data-driven methods and established risk assessment models. While prior research has addressed various aspects of heritage fire risks, these efforts are typically fragmented and separately capture partial dimensions of risks. The lack of integration may result in incomprehensive assessments that weaken the reliability of results.
(2) The limited integration is further reflected in the scarce research into developing standardized approaches for data requirement identification, risk modeling, and automated data extraction and processing. Although BIM and GIS contain vast amounts of raw data, there is limited understanding of which data elements are most relevant to fire risk assessment and how they can be systematically converted into quantitative risk indicators. These issues are the key bottlenecks that influence efficient data-driven risk assessments.
(3) Challenges also arise from the heritage buildings, which often possess irregular architectural geometries, non-standard material compositions, and strict conservation constraints, thereby complicating the application of data-driven risk assessment methods.

Given the afore-mentioned limitations, this study aims to propose a new fire risk assessment model for historical centers, RATER, as well as develop a BIM and GIS-integrated system that comprehensively incorporates various risk factors and facilitates risk level analysis for heritage buildings. The four main objectives of this study include: (1) to develop RATER that identifies and categorizes fire risk factors across fire hazards, heritage and firefighting equipment, surrounding environment, and safety management dimensions, (2) to determine risk factor weightings through AHP-based analysis to reveal consensus and divergence in risk perceptions across four professional groups, (3) to establish a BIM-GIS integrated system for 3D spatial fire risk visualization and decision support, and (4) to validate the applicability of RATER through cross-comparison assessments on three real-world heritage buildings.

## 3. Methodology

This study employs mixed methods to develop and validate RATER through four phases. In the RATER model, the fire risk of a

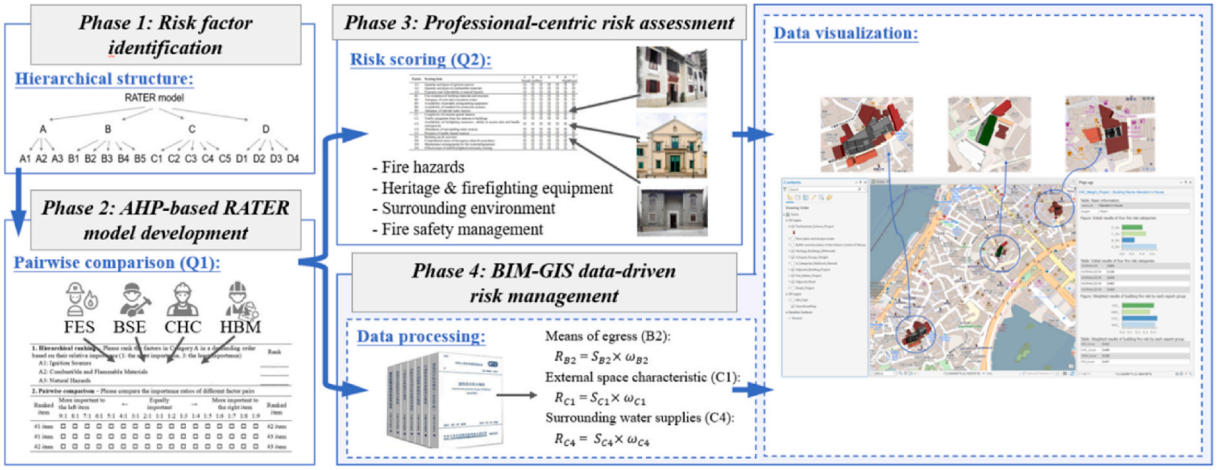



**Fig. 1.** Development workflow of the proposed fire risk assessment model.

heritage building is quantified using a composite risk index, which integrates multiple factors according to their weights and scores. The factor weight reflects the relative importance of each risk factor generically, while the factor score quantifies the site-specific condition of each factor. Correspondingly, the four development and validation phases align with the composition of this risk index. Phase 1 involves the identification and categorization of fire risk factors through an extensive literature review to establish the overall structure of the RATER model. Next, in Phase 2, the generic factor weights of the RATER model are determined through multi-stakeholder pairwise comparisons using the AHP approach. Subsequently, this study employs two independent assessment approaches to determine the factor scores. In Phase 3, factor scores are derived from a professional-centric assessment to capture practitioner judgement. In parallel, a data-driven assessment is also conducted to compute factor scores objectively using spatial and geometric attributes from BIM models and GIS maps, as shown in Phase 4. These two independent assessment approaches are then cross-validated to examine the consistency of results. Eventually, the assessment results are imported and visualized in a BIM-GIS integrated system. Fig. 1 illustrates four-phase development and validation processes. Additionally, the calculation procedures for weight determination and score evaluation showcased in Phases 2 to 4 are illustrated in Appendix A, Fig. A.

### 3.1. Risk factor identification

To systematically identify fire risk factors affecting heritage buildings, this study conducts an extensive literature review to incorporate domain knowledge from a wide range of disciplines, including fire safety engineering, cultural heritage conservation, emergency response, and building management. Following the PRISMA guidelines (see Fig. 2) [47], the literature selection process comprises four stages: identification, screening, eligibility assessment, and final inclusion. In the identification stage, peer-reviewed articles are searched in four academic databases, i.e., Scopus, Web of Science, ScienceDirect, and CNKI. In addition, technical reports, case studies of fire incidents, and safety compliance and regulations in heritage buildings are collected from relevant authorities, including UNESCO [48], ICOMOS [49], China Standard [50], Historic England [51], and USA NFPA [52]. The inclusion and exclusion criteria are as follows: (1) only studies addressing fire risk or fire safety in heritage buildings are included; (2) only studies pertaining to risk factors or assessment methodologies are included; and (3) studies focusing solely on wildland-urban interface fires without building-specific contexts are excluded. As a result, 253 records are retrieved after removing the duplicates. Subsequently, manual screening based on titles, abstracts, and main contents is conducted to refine the set of records. A total of 22 records is retained as the basis to identify relevant fire risk factors and form the RATER model. Since these 22 documents are generic and are from diverse heritage and fire safety sources, the identified factors capture fundamental risk dimensions rather than region-specific regulatory requirements, thereby ensuring the generalizability of the RATER model.

Based on the identified documents, a synthesis and categorization process is then carried out to merge similar factor concepts into more generalizable representations and ensure conceptual completeness. Specifically, the categorization is inspired by the IRGC Risk Governance Framework [53] and the source-pathway-receptor model [48,54]. It follows the causal and control progression of heritage fires from its source, through the pathway, to the affectors. The source represents where the hazard originates. The pathway describes how the hazard develops, propagates, and interacts with control measures, both internally and externally. The affectors refer to the entities or conditions that can either be harmed by or help mitigate the hazard. In line with this risk causality and control logic, RATER is categorized to reflect and measure the potential fire risks from fire hazards, heritage and firefighting equipment, surrounding environment, and fire safety management, with the identified factors shown in Table 2. Moreover, the factors in RATER are cross-checked with prior fire risk frameworks to ensure comprehensive coverage and minimize redundancy.

First, *fire hazards* (Category A) refer to factors that significantly increase the likelihood or severity of a fire, considering direct threats such as *ignition sources* (A1), indirect threats like *combustible and flammable materials* (A2), and incidental threats such as *natural hazards* (A3). Second, *heritage and firefighting equipment* (Category B) encompasses the features of buildings and firefighting services that affect fire propagation, emergency evacuation, and fire suppression. Correspondingly, *building inherent features* (B1), *means of egress* (B2), *portable firefighting devices* (B3), *integrated fire service installations* (B4), and *water supplies* (B5) are incorporated in this category. Third, *surrounding environment* (Category C) relates to external conditions that influence the severity and response actions for fires in heritage buildings and historic centers, such as *external space characteristics* (C1), *road traffic* (C2), *firefighting resources* (C3), *surrounding water supplies* (C4), and *surrounding potential dangers* (C5). Last, *fire safety management* (Category D) incorporates the plans and practices among stakeholders in order to reduce the risks and impacts of fires in heritage buildings, such as *occupancy status* (D1), *emergency plans* (D2), *routine maintenance* (D3), and *multi-stakeholder training* (D4). Consequently, the RATER includes a total of 17 risk factors across four categories.

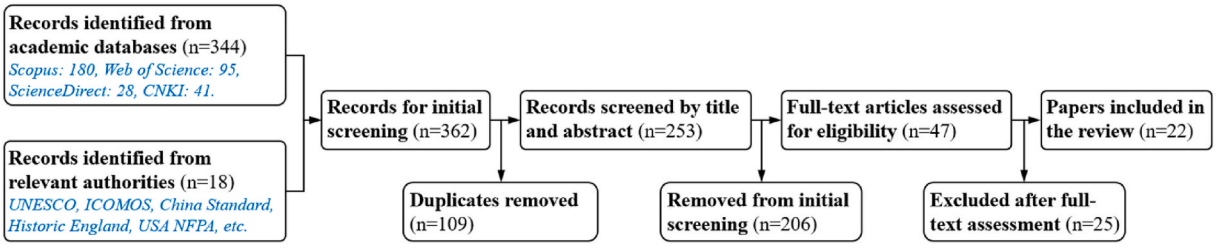



**Fig. 2.** Literature review process for fire risk factor identification.

**Table 2**
Fire risk factors in the RATER.

| Factor | Description | Sources |
|---|---|---|
| ***A: Fire hazards*** | | |
| *A1: Ignition sources* | Agents that produce sparks, flames, or heat to initiate combustion, including naked lights, electrical faults, and hot surfaces. | [55,56] |
| *A2: Combustible and flammable materials* | Combustible and flammable materials are items that can easily ignite or burn to sustain combustion when exposed to an ignition source, such as furniture and furnishings, wood, textiles, plastics, and gas systems. | [57,58] |
| *A3: Natural hazards* | Natural events that increase the likelihood or severity of fires in heritage buildings, such as lightning strikes, earthquake-induced ignitions, and extreme climatic conditions (e.g., typhoons, droughts, and heatwaves). | [59] |
| ***B: Heritage and firefighting equipment*** | | |
| *B1: Building inherent features* | Building attributes that affect the fire spread and propagation, such as fire resistance, building safety layouts (e.g., fire compartmentation and separation) and functional areas (e.g., hazardous storage areas). | [60,61] |
| *B2: Means of egress* | Means of egress are the pathways and exits designed to assist occupants and firefighters in escape and rescue operations. | [62,63] |
| *B3: Portable firefighting devices* | Apparatuses that can be easily carried to combat fires in heritage buildings, e.g., fire extinguishers and blankets. | [64] |
| *B4: Integrated fire service installations* | Fixed systems installed within heritage buildings that assist occupants and firefighters in fire response, including alarms, sensors, sprinkler systems, gas flooding systems, and emergency electricity and lighting systems. | [64,65] |
| *B5: Water supplies* | Sources and apparatuses designed to supply water with sufficient pressures and flow rates, e.g., pipe systems, tanks and pumps. | [66] |
| ***C: Surrounding environment*** | | |
| *C1: External space characteristics* | Spatial conditions that determine the exposure and isolation of heritage buildings and affect the spread and severity of fires, such as fire separation distance, frontage, geographical elevations, and adjacent structures. | [67] |
| *C2: Road traffic* | Transportation situations that affect the accessibility and mobility of first responders and fire engines toward the affected heritage buildings, covering static road information and dynamic traffic flow information. | [68,69] |
| *C3: Firefighting resources* | Facilities, engines, and human resources allocated for fire responses in heritage buildings. | [70,71] |
| *C4: Surrounding water supplies* | Sources of water available in the vicinity of heritage buildings that can be used to combat fires, such as fire hydrants and natural water bodies. | [66] |
| *C5: Surrounding potential dangers* | Adverse elements in the vicinity of heritage buildings that may affect the severity of fires, such as gas stations, vegetation, and power lines. | [72] |
| ***D: Fire safety management*** | | |
| *D1: Occupancy status* | Occupancy conditions where the heritage building is generally occupied and used, including regular (e.g., tourism), occasional (e.g., cultural events), and temporary activities (e.g., renovation work). | [73] |
| *D2: Emergency plans* | Protocols that regulate the roles, responsibilities, procedures, and resources of different emergency participants involved in preventing, preparing, and responding to fire incidents in heritages. | [74] |
| *D3: Routine maintenance* | Activities of inspecting, testing, repairing, and replacing heritage structures and firefighting equipment to ensure functionality against fire incidents in heritage buildings. | [75] |
| *D4: Multi-stakeholder training* | Joint efforts that involve various parties in training activities to practice fire responses in heritage buildings. | [76] |

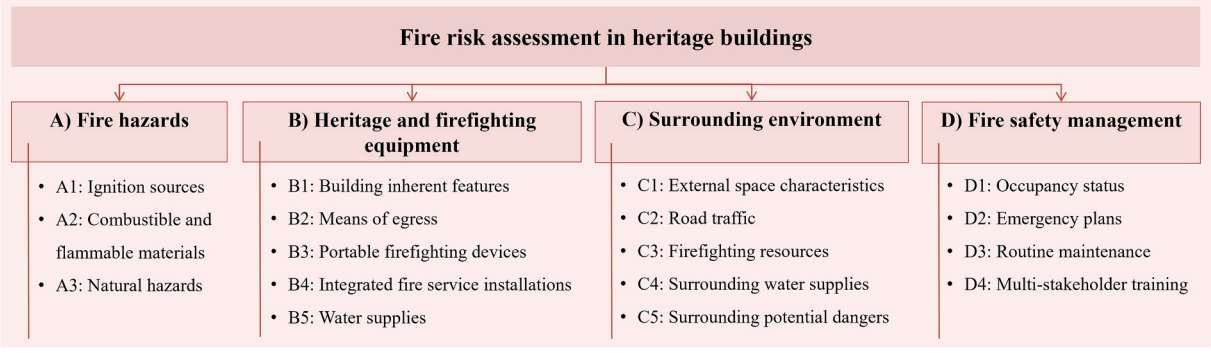



**Fig. 3.** Hierarchical structure of RATER.

### *3.2. AHP-based RATER development*

Next, RATER development is to measure the relative importance of different risk factors by absorbing professionals' judgments from different fields. According to UNESCO's Fire Risk Management Guide [48], effective fire risk management in heritage buildings requires broad stakeholder collaboration. However, not all of the stakeholders have equal influence on fire-risk factor weighting. Based on a stakeholder salience analysis, this study identifies four core professional groups most directly relevant to heritage fire risk assessments, i.e., fire emergency services (FES), heritage building management (HBM), cultural heritage conservation (CHC), and building safety engineering (BSE). These four groups form the professional panel to assess and compare the relative importance of factor pairs based on their domain knowledge and first-hand experience. The detailed composition and sample size of each group are provided in Section 4.2. To help assess the relative importance of different risk factors, the AHP method is employed in this study.

Correspondingly, the proposed RATER is structured in a three-level hierarchical manner, as shown in Fig. 3.

Subsequently, pairwise comparisons are conducted by the professional panel members to determine the weights of the factors in RATER. Structured questionnaires (Q1) are designed to perform pairwise comparisons at two levels: (1) among different factors within each category and (2) among the categories.

To ensure smooth pairwise comparisons in practice, pilot testing was conducted with six researchers and professionals in relevant backgrounds to refine factor definitions and verify the comparison workflow. The pilot test revealed that understanding factor terminologies and maintaining logical consistency were the two primary challenges. To address the first challenge, a pre-survey introduction is provided to respondents with clear descriptions and visual references to help form a shared understanding of factor definitions. For logical consistency, the official questions in Q1 are structured in two sequential steps. First, respondents are asked to rank factors within each category in a descending order of importance. Second, pairwise comparisons are conducted based on the ranking order, which naturally constrains comparison ratios and reduces potential inconsistencies. Afterward, respondents are also instructed to justify their ratings. This helps double-check consistency and provides additional insights into their prioritization of fire risks. For each comparison, the pairwise relative weighting is determined based on a standard 9-point scale [77], where 1 indicates equal importance and 9 indicates extreme importance of one factor over the counterpart. The full pairwise comparison sheet for risk factors is provided in Appendix B.

After conducting pairwise comparisons, it is important to examine the logical consistency of the results. This study adopts the concepts of consistency index (CI) and consistency ratio (CR) as the statistical measures to assess the effectiveness of professionals' judgments [77]. The CI quantifies the degree of consistency in the pairwise comparison matrix $\boldsymbol{A}$. The CR value is calculated by comparing the CI to the random index (RI), indicating the consistency level of the pairwise judgments against random chance.

Next, weights are calculated to derive the relative importance of each fire risk factor. For an individual professional, the pairwise comparison matrix $\boldsymbol{A}$ is normalized to ensure that all values are on a comparable scale (Equation (1)). Then, relative weights are obtained by averaging each $i$-th row item (Equation (2)).

$$a'_{ij} = a_{ij} \Big/ \left( \sum\nolimits_{i=1}^{n} a_{ij} \right) \tag{1}$$

$$\omega_{k,i}^{ind} = \left( \sum\nolimits_{j=1}^{n} a'_{ij} \right) \Big/ n \tag{2}$$

where $a'_{ij}$ represents the normalized value of $a_{ij}$ according to the $j$-th column in the comparison matrix, $\omega_{k,i}^{ind}$ denotes the individual weight of factor $i$ assessed by $k$-th professional within the professional group $g$. Also, $n$ denotes the total number of factors being compared within the corresponding level or category.

To combine individual weights into a professional group weight, the geometric mean calculation and normalization are used to balance professional consensus [78] as shown in Equations (3) and (4).

$$\omega_{g,i}^{grp} = \left( \prod\nolimits_{k=1}^{m_g} \omega_{k,i}^{ind} \right)^{1/m_g} \tag{3}$$

$$\omega_{g,i}^{norm} = \omega_{g,i}^{grp} \Big/ \left( \sum\nolimits_{i=1}^{n} \omega_{g,i}^{grp} \right) \tag{4}$$

where $\omega_{g,i}^{grp}$ represents the group weight in terms of the $i$-th factor and the $g$-th professional group, $\omega_{k,i}^{ind}$ is the weight assigned to the $i$-th factor by the $k$-th professional, $m_g$ denotes the number of participants in the professional group $g$, and $\omega_{g,i}^{norm}$ is the group weight after normalization.

Given that RATER is a two-level hierarchical model, including a category level (L1) and a factor level (L2), their corresponding group weights $\omega_{g,i}^{norm,L1}$, $\omega_{g,i}^{norm,L2}$ are calculated using Equations (1)–(4). Traditional AHP assumes that category weights are fully derived from factor weights in a strict hierarchy. In contrast, RATER clusters factors into groups, treating categories and factors as related but not rigidly stratified. The ultimate goal remains to ascertain the relative importance of individual factors, with categorization functioning merely as a practical framework for simplifying comparisons. Therefore, a weight adjustment coefficient $K_{g,i}$ is introduced to balance out any disproportion that may arise when categories contain different numbers of factors, as shown in Equation (5). The coefficient $K_{g,i}$ consists of two parts. The first term $(n/N)$ compensates for differences in category size by scaling each category's contribution according to its share of all factors. This prevents categories with fewer factors from being over-represented and those with more factors from being under-represented in the global weighting. The second term $\left[ N / \sum_{j=1}^{C} \left( \omega_{g,i}^{norm,L1} \times n \right) \right]$ provides global normalization so that the total adjusted weights sum to one. Collectively, these terms eliminate size-induced distortion and maintain consistent comparability of factor weights across categories. Consequently, the global weight $\omega_{g,i}^{glob}$ of each factor for different groups is calculated using Equation (6).

$$K_{g,i} = (n / N) \times \left[ N \Big/ \sum_{j=1}^{C} \left( \omega_{g,i}^{norm,L1} \times n \right) \right] \tag{5}$$

$$\omega_{g,i}^{glob} = \omega_{g,i}^{norm,L1} \times \omega_{g,i}^{norm,L2} \times K_{g,i} \tag{6}$$

where $n$ is the number of factors within the $j$-th category, $N$ is the total number of factors, $C$ is the total number of categories.

To obtain holistic perspectives from all four professional groups, the global weights $\omega_{g,i}^{glob}$ of each group are averaged to derive an integrated weight ($\omega_i^{norm}$) for each factor using Equation (7). This integrated weight is then applied in subsequent risk analysis across all heritage sites.

$$\omega_i^{norm} = \left( \sum_{g=1}^{G} \omega_{g,i}^{glob} \right) \Big/ G \tag{7}$$

where $G$ is the number of professional groups, $\omega_i^{norm}$ is the integrated weight of factor $i$, and $\omega_{g,i}^{glob}$ is the global weight of the professional group $g$.

To check consistency across respondents, three inter-rater reliability analyses are conducted: the AHP consensus indicator ($S^*$) and relative homogeneity index ($S$), Cronbach's alpha, and Kendall's coefficient of concordance ($W$). The AHP consensus indicator and relative homogeneity index are used to assess group-level agreement among professionals' pairwise comparisons, with values between 0.75 and 0.85 indicating high consensus and values above 0.85 indicating very high consensus [79]. Cronbach's alpha is calculated to examine the internal consistency reliability of the judgments, where values above 0.70 are considered acceptable and values above 0.80 are regarded as good [80]. Kendall's coefficient of concordance ($W$) is used to evaluate rank-order agreement among professionals. $W$ ranges from 0 (no agreement) to 1 (perfect agreement), and its associated probability value ($p < 0.05$) indicates the statistically significant concordance [81].

Sensitivity analysis is also conducted to assess the robustness of factor importance rankings. Each factor weight is perturbed by ± 10%, ±20%, and ±30% while maintaining the total weight constant. For each scenario, the adjusted factor weights are computed, and the relative importance rankings of all factors are reassessed. Ranking stability is further examined using Spearman's rank correlation coefficient ($\rho$), with values exceeding 0.90 indicating very high stability.

Apart from the rating of relative importance, Q1 also collects qualitative feedback from the professionals. Multiple-choice questions are employed to investigate the current status of heritage risk assessments in terms of adequacy and factor consideration. Moreover, the challenges in existing practices are identified. Open-ended comments are collected to gather the professionals' insights into the underlying reasons for their judgments and potential improvement measures. This helps interpret the results and complement the AHP weighting analysis. As a result, their responses are transcribed and clustered to present the representative viewpoints of different roles in the subsequent analysis stage.

### 3.3. Professional-centric risk assessment

Based on the weighted RATER, this study further conducts both professional-involved surveys and data-driven analysis to assess the fire risks of real-world heritage buildings. The professional-centric risk assessment intends to reveal deeper, more detailed insights from practitioners about the conditions of the heritage buildings, further forming collective knowledge to measure their fire risks. To do so, a field survey is employed to collect data from on-site practitioners. The selection criteria of the survey participants are based on their familiarity level with the heritage sites as well as their work duties, requiring them to have either prior experience or current involvement in the heritage buildings, along with relevant management and safety responsibilities.

To structure the data collection process, a questionnaire (Q2) for heritage risk assessment (as shown in Appendix C) is devised, which asks participants to evaluate risk scores of the 17 factors independently based on the current status of heritage buildings. A set of 7-point Likert-scale questions (1 = very low risk, 7 = very high risk) is utilized to collect the responses of the participants.

The responses of Q2 are then analyzed to calculate the risk scores of the 17 influencing factors. The rating results of the Likert-scale questions are first normalized on a 0 to 100 scale, as shown in Equation (8). Next, the average risk score ($S_{p,i}^{norm}$) of each factor among different participants on each heritage site is then calculated using Equation (9).

$$S_{u,i}^{ind} = (L_{u,i} - 1) / 6 \times 100 \tag{8}$$

$$S_{p,i}^{norm} = \left( \sum_{u=1}^{m_p} S_{u,i}^{ind} \right) \Big/ m_p \tag{9}$$

where $L_{u,i}$ is the original Likert scire (1–7) assigned by the $u$-th participant assessed to the $i$-th factor; $S_{u,i}^{ind}$ is the normalized individual score (0–100); $S_{p,i}^{norm}$ is the average risk score; $m_p$ is the number of participants on the heritage site $p$.

Subsequently, this risk score $S_{p,i}^{norm}$ is combined with the integrated weight ($\omega_i^{norm}$) to calculate the risk index $R_{p,i}$ of each factor, as shown in Equation (11). The total fire risk index $R_p$ is obtained by summing all $R_{p,i}$ as in Equation (10). After calculating the risk index

of each professional group, the final integrated result is obtained by averaging across the four groups, assuming their perspectives are equally important for fire risk assessment.

$$R_p = \sum_{i=1}^{N} R_{p,i} \tag{10}$$

$$R_{p,i} = S_{p,i}^{norm} \times \omega_i^{norm} \tag{11}$$

where $S_{p,i}^{norm}$ is the average group score, $\omega_i^{norm}$ is the integrated weight of factor $i$; $R_{p,i}$ is the risk index of the $p$-th heritage site regarding the $i$-th factor, and $R_p$ is the risk index of the $p$-th heritage site considering all factors.

The weighted-average formulation of the risk index in Equations (8)–(11) inherently introduces a compensatory effect, whereby low scores in certain factors may be partially offset by higher scores in others. To address this limitation, the unified risk index $R_p$ is reported together with the full set of individual factor scores $S_{p,i}^{norm}$ and their corresponding weights [26]. These individual factor results enable practitioners to identify critical, non-compensable weaknesses that should not be obscured by the aggregated score.

The calculated risk index is expressed on a 0–100 scale. To enhance practical usability, this study further classifies the index scale into five indicative levels, i.e., very low (0–20), low (21–40), moderate (41–60), high (61–80), and very high (81–100). This classification helps practitioners better interpret the relative severity of risks, both across different buildings and across individual factors, and support the prioritization of mitigation actions accordingly.

### 3.4. BIM and GIS-integrated data-driven risk management

Apart from the professional-centric assessment, this study further develops a BIM and GIS-integrated system for data-driven risk management. Specifically, the system serves two purposes: (1) it utilizes existing BIM and GIS data to conduct quantitative risk analysis, forming an objective alternative to supplement and cross-check the professional-centric assessment, and (2) it visualizes the geospatial distribution of fire risk results across heritage buildings to facilitate informed decision-making in fire risk management. The development of the integrated system incorporates two critical parts, i.e., risk formulation as well as data processing and visualization.

#### 3.4.1. Risk formulation for data-driven score generation

For risk formulation, the factors identified in RATER are structured into mathematical formulas for quantitative and objective fire risk assessment. Ideally, risk formulation should be applied to all factors. Whilst in practice, challenges such as insufficient and inaccurate data, as well as the absence of standardized regulatory codes, exist and restrict the full implementation. As an alternative, the present study prioritizes factors that are readily accessible via BIM models and GIS systems as exemplars to demonstrate the potential of data-driven risk management using RATER. Consequently, factors B2, C1, and C4 are selected to conduct risk formulations. Existing standards, practical guidelines, and academic articles are used as references to establish quantitative formulas for individual risk factors. Although the current implementation includes only three factors, it remains methodologically relevant because it establishes risk formulation and computational data analysis workflow to integrate RATER within a BIM-GIS environment, forming a scalable foundation for future extension when more complete datasets become accessible.

For *means of egress* (B2), its quantitative risk model $S_{B2}$ focuses on the calculation of evacuation time according to the regulations [11,82–85]. Hence, the risk model compares the required evacuation travel time ($T_{req}$) and the available safety time ($T_{avl}$) to determine whether the evacuation process can be completed safely within the allowable time, as shown in Equations (12) and (13).

$$T_{req} = L_{max} / V_{evac} \tag{12}$$

$$S_{B2} = \begin{cases} \left(T_{req}/T_{avl}\right) \times 100, T_{req} < T_{avl} \\ 100, T_{req} \geq T_{avl} \end{cases} \tag{13}$$

where $L_{max}$ represents the largest length of the evacuation path (in meters), $V_{evac}$ is the evacuation speed, $T_{req}$ represents the evacuation travel time (the movement component of total evacuation time), $T_{avl}$ denotes the available safety time, and $S_{B2}$ is the risk score of factor B2, ranging from 0 (lowest risk) to 100 (highest risk). Suggested values for $V_{evac}$ and $T_{avl}$ are 1.2 m/s for an ordinary crowd [86and90] seconds as regulated in Ref. [88]. These values serve as default reference parameters and can be replaced with more precise inputs when heritage-specific data are available.

*External space characteristics* (C1) integrate proximity accessibility and spatial separation to assess the degree of exposure and isolation of heritage buildings during fire incidents (Equation (14)):

$$S_{C1} = \varphi_1 S_A + \varphi_2 S_S \tag{14}$$

where $\varphi_1$ and $\varphi_2$ are coefficients representing the relative importance of proximity accessibility $S_A$ and spatial separation $S_S$, assumed to be equally important by default.

Proximity accessibility $S_A$ evaluates whether $s$-th adjacent road width ($W_{adj,s}$) meets emergency vehicle requirement (i.e., $W_{min} = 4$ meters in Ref. [89]). In case no suitable roads exist, $S_A$ further considers the width of other potential roads ($W_{alt}$) within a designated vicinity area (i.e., 30 m in Ref. [50]) as alternative parking options, as shown in Equation (15):

$$S_A = \begin{cases} \sum_{s=1}^{N_{road}} 1_{\{W_{adj,s} < W_{min}\}} \Big/ N_{road} \times 100, \text{if } \exists W_{adj,s} \geq W_{min} \\ 87.5, \text{if } \forall W_{adj} < W_{min} \text{ and } \exists W_{alt} \geq W_{min} \\ 100, \text{if } \forall (W_{adj}, W_{alt}) < W_{min} \end{cases} \tag{15}$$

where $N_{road}$ represent the number of directly adjacent roads.

The spatial separation $S_S$ assesses the actual distances $d_i$ between the heritage and its adjacent buildings against the minimum regulated fire separation distance $d_{i,min}$ for the sake of fire propagation prevention (Equation (16)):

$$S_S = \sum_{i=1}^{N_b} (1 - d_i/d_{i,min})_{\{d_i < d_{i,min}\}} \times 100 \tag{16}$$

*Surrounding water supplies* (C4) focuses on the availability and accessibility of water sources required for firefighting in terms of quantity ($S_N$), relative location ($S_D$), and coverage ($S_C$) according to Refs. [66,68] (Equation (17)).

$$S_{C4} = \mu_1 S_N + \mu_2 S_D + \mu_3 S_C \tag{17}$$

where $\mu_1$ to $\mu_3$ are the weighting coefficients and are set by default to be evenly distributed.

Specifically, $S_N$ compares the actual number of hydrants ($N_{h,a}$) and the required number of hydrants ($N_{h,r}$), which is 2 for street-facing sides or 1 for non-street-facing sides [66] (Equation (18)).

$$S_N = \begin{cases} (1 - N_{h,act}/N_{h,req}) \times 100, \text{if } N_{h,act} < N_{h,req} \\ 0, \text{if } N_{h,act} \geq N_{h,req} \end{cases} \tag{18}$$

The relative location ($S_D$) evaluates whether hydrants are positioned at appropriate distances $D_h$ from building walls by comparing with the optimal range from $D_{h,low}$ (5 m) to $D_{h,up}$ (40 m) and the maximum limit $D_{h,max}$ (150 m) [66] (Equation (19) and (20)).

$$S_D = \left( \sum_{i=1}^{N_{h,act}} S_{D,i} \right) \Big/ N_{h,act} \tag{19}$$

$$S_{D,i} = \begin{cases} (D_{h,low} - D_{h,i})/D_{h,low} \times 100, \text{if } 0 \leq D_{h,i} < D_{h,low} \\ 0, \text{if } D_{h,low} \leq D_{h,i} < D_{h,up} \\ (D_{h,i} - D_{h,up})/(D_{h,max} - D_{h,up}) \times 100, \text{if } D_{h,up} \leq D_{h,i} < D_{h,max} \\ 100, \text{if } D_{h,i} \geq D_{h,max} \end{cases} \tag{20}$$

The coverage ($S_C$) considers the length of the building perimeter ($L_{cov}$) covered within the hydrant's effective radius (150 m [66])

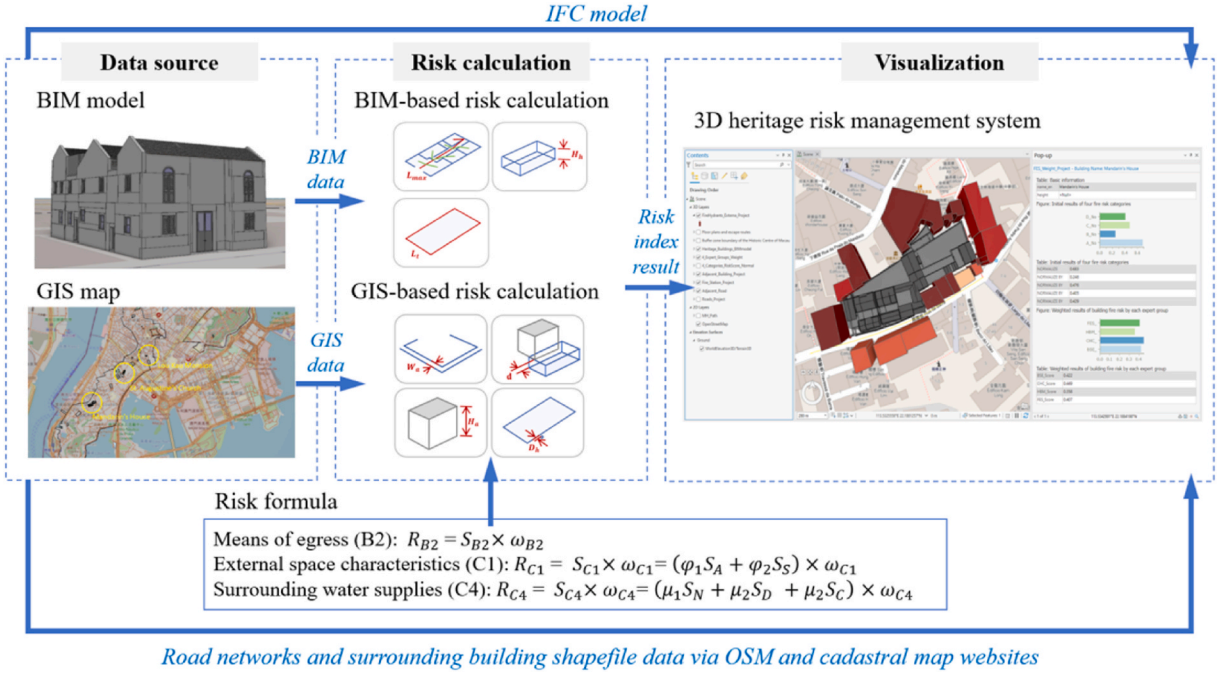



**Fig. 4.** BIM-GIS data integration workflow.

against the total perimeter length ($L_{tot}$) (Equation (21)).

$$S_C = (1 - L_{cov} / L_{tot}) \times 100 \tag{21}$$

From Equations (12)–(21), the risk scores of the three factors are obtained. To compute the overall risk index, these scores are integrated with the AHP-derived factor weights using the same formations (i.e., Equations (10) and (11)) as in the professional-centric assessment. The only difference is that the professional-rated scores are substituted by the data-driven risk scores.

### 3.4.2. *Data processing and visualization*

After formulating the risk models, it is critical to acquire actual heritage data to conduct risk calculations. Specifically, BIM models provide geometric and semantic information of heritage buildings, while GIS systems offer geo-spatial data of surrounding environments. Therefore, open-source BIM data schema and GIS map databases, i.e., IFC and OpenStreetMap (OSM), are utilized in this study.

For *means of egress* (B2), it requires the calculation of the largest evacuation path distance using the heritage floorplan. Correspondingly, the 2D geometries of the heritage building are extracted from the BIM model using Python scripts and IfcOpenShell [1] to compute the spatial accessibility graph, including the central locations of rooms and doors as nodes and evacuation paths as edges. Self-developed scripts are utilized to extract paths from rooms to doors and from doors to exits within the BIM floorplan to compute the longest evacuation path via iterative searching.

For *external space characteristics* (C1), this study extracts geo-spatial data from OSM and the Macao SAR Directorate of Cartography and Cadaster Services database [87], and combines it with BIM data for risk analysis. Regarding proximity accessibility, the geo-location coordinates and width of the surrounding roads are directly obtained from OSM to examine emergency vehicular access. With respect to spatial separation, geospatial data in GIS maps, including building heights, locations, and footprint boundaries, are extracted to calculate the actual distance and the regulated minimum distance between the heritage and the adjacent buildings.

For *surrounding water supplies* (C4), the information sources are from both BIM models and GIS maps. First, hydrant location data is acquired from the GIS map database to count the numbers and calculate their spacings. Next, the footprint boundary of the heritage is derived from BIM geometries, which are further utilized to analyze the perpendicular distances between hydrants and building exterior walls. Last, the footprint boundary of the heritage is overlaid with the hydrants' effective circular area to determine the covered segments of the perimeter.

In addition to information provision, BIM models and GIS maps also provide foundational 3D models for data visualization in the integrated system. The current implementation of the BIM-GIS system is developed as an ArcGIS project with file-based data storage, functioning as the data management and presentation environment. IFC models are first imported into ArcGIS using a built-in data interoperability function [90], which aligns the building models with OSM footprints through WGS84 transformation. GIS meshes, including adjacent buildings, roads, and fire hydrants, are extracted from OSM and converted into GeoJSON or Shapefile formats for system import. Regarding data processing, risk calculation data is also converted into compatible attribute tables and linked to corresponding spatial entities within ArcGIS. All processed data is then re-integrated into ArcGIS for holistic visualization. Fig. 4 shows the data integration workflow for the BIM-GIS system development.

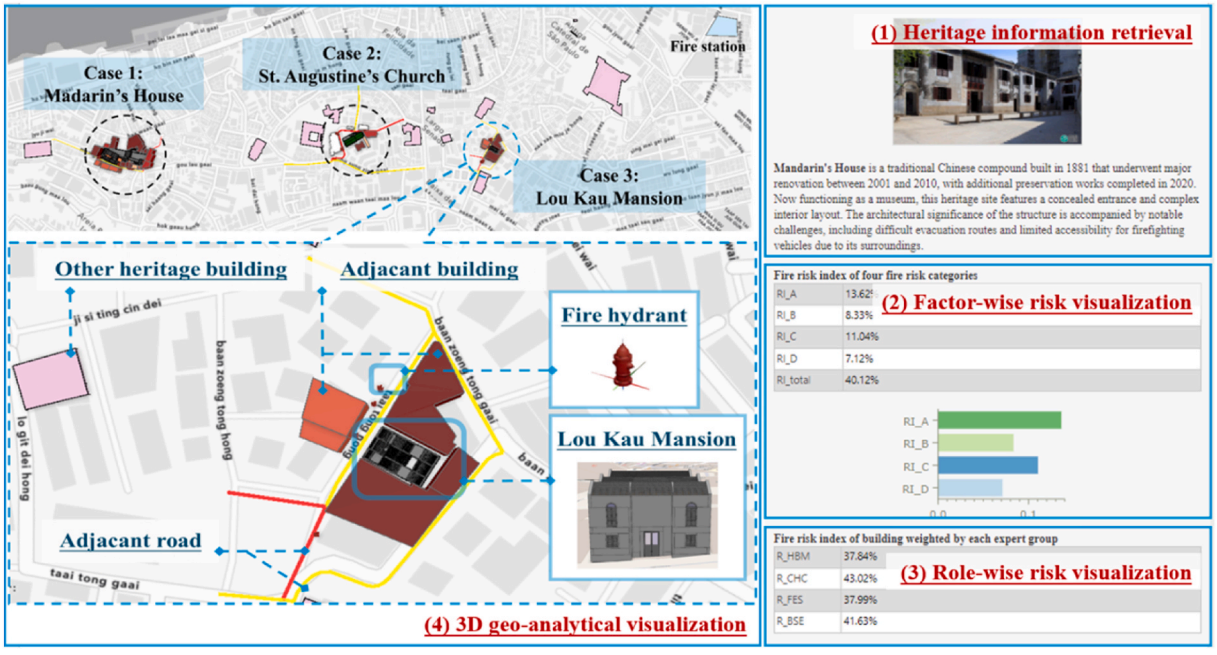



**Fig. 5.** Functions and interfaces of the integrated system.

The system has four critical functions, i.e., heritage information retrieval, factor-wise risk visualization, role-wise risk visualization, and 3D geo-analytical visualization (Fig. 5). Information retrieval enables users to access essential heritage data such as names, physical characteristics, historical significance, and risk indices to gain a better understanding of the building status. Role-wise risk visualization presents the risk rating from different professional groups using charts to vividly show the similarities and differences between their considerations and judgments. Factor-wise risk visualization illustrates the quantitative results of both professional-centric and data-driven risk assessments to aid stakeholders in interpreting the present conditions of heritage buildings against potential fire incidents. The 3D geo-analytical visualization integrates 3D building models and geospatial meshes to visualize details of building internal and external characteristics for fire risk management. The system includes BIM layers for internal building characteristics, GIS mesh layers for surrounding structures (i.e., adjacent buildings, fire stations, and other heritages in the historic center), and GIS road network layers for accessibility analysis. These functions enable stakeholders to identify critical weaknesses in current fire safety practices while facilitating informed decision-making for enhanced protection measures.

## 4. Real-world implementation with case studies

### 4.1. Background of case studies

Three heritage buildings in Macao SAR serve as case studies: St. Augustine's Church, Lou Kau Mansion and Mandarin's House. These buildings are integral components of the Historic Centre of Macao SAR in the UNESCO World Heritage Site, with various architectural styles, historical periods, and functionalities. Table 3 summarizes their background and characteristics.

### 4.2. AHP-based risk weight analysis

To assess the relative importance of fire risk factors, professionals were recruited through purposive and snowball sampling. Ethical approval for this study was obtained from the panel on research ethics of the University of Macao (ethics assessment ID: SSHRE23-APP089-FST) before the commencement of the survey. Participant recruitment was conducted through purposive and snowball sampling in collaboration with relevant professional bodies and institutions in Macao and the surrounding regions. The FES group consisted of 10 fire department officers and firefighters from the Macao Fire Services Bureau and the Hong Kong Institution of Engineers (Fire Division), all with hands-on experience in on-site fire response. The HBM group comprised 12 heritage safety inspectors and asset managers recruited through field surveys in the Historic Centre of Macao, each familiar with the daily operations and potential risks of heritage buildings. The CHC group included 15 conservation and restoration professionals from the Macao Cultural Affairs Bureau (Department of Cultural Heritage) and the Shaanxi Provincial Institute of Cultural Heritage. Finally, the BSE group comprised 16 construction supervision engineers and fire protection engineers contacted through the Macao Institute of Engineers and other professional networks, all possessing expertise in fire system design and safety assessment. The sample size for each group was determined with reference to relevant regulations [60,64,91] and prior studies [30–35] to ensure both disciplinary diversity and sufficient representation within each professional domain. Table 4 summarizes the demographic information of the 53 participants in the professional panel.

The responses were analyzed, and both individual and group-level reliability metrics were computed to validate the consistency of pairwise comparison results. At the individual level, all 53 matrices achieved consistency ratios below 0.1, confirming logical consistency. At the group level, the inter-rater reliability analysis demonstrated a high level of consistency across all professional groups: (1) The AHP consensus indicator ($S^{*}$) ranged from 0.851 to 0.914, and the relative homogeneity indices ($S$) ranged from 0.865 to 0.922. These results indicate strong inter-rater consensus among individual groups. (2) Cronbach's alpha values ranged from 0.755 (CHC) to 0.928 (HBM), suggesting acceptable to high internal consistency and reliability among the professionals' judgments. Additionally, Kendall's coefficients of concordance ($W$) were calculated as a supplementary metric. The $W$ values ranged from 0.221 to 0.598, indicating low to moderate rank-order agreement. In other words, members of the same professional group generally share coherent views on fire-risk importance. Nevertheless, some intra-group variation persists, reflecting the diversity of individual professional experience and judgment. The complete inter-rater reliability statistics are provided in Appendix D.2. Then, the weight comparison results of 17 fire risk factors by the four professional groups were calculated using equations in Section 3.2, as shown in

**Table 3**
Overview of the three heritage sites in the case studies.

| Site | St. Augustine's Church | Lou Kau Mansion | Mandarin's House |
|---|---|---|---|
| **Historical background** | Built in 1591, major reconstruction in 1875, and weather damage repairs in 2016 | Built in 1889, major restoration in 2007, fire caused by nearby construction in 2016 | Built in 1881, major renovation in 2001–2010, preservation works in 2020 |
| **Architectural style** | Classical Renaissance-style religious edifice | Chinese courtyard-style residence | Chinese courtyard-style residence |
| **Structural type** | Brick and wood | Blue bricks and wood | Blue bricks and wood |
| **Function** | Church, with occasional religious activities | Exhibition building with a tourism function | Exhibition building with a tourism function |
| **Surroundings** | Located in a dense historic area, with limited emergency access | Located in a popular tourist area, with many shops and visitors | Located in a mixed-use neighborhood with both commercial and residential properties |
| **Key challenges** | Large crowds during religious holidays, restricted firefighting and evacuation | Compact block layout, limited evacuation routes | Concealed entrance, complex interior layout, restricted access for firefighting vehicles |

**Table 4**
Demographic information of Q1 survey participants across four professional groups.

| | Number of responses | | | | |
|---|---|---|---|---|---|
| | FES[a] (total = 10) | HBM[a] (total = 12) | CHC[a] (total = 15) | BSE[a] (total = 16) | Sum (total = 53) |
| (1) Experience | | | | | |
| Less than 1 year | 5 | 5 | 0 | 2 | 12 |
| 1–5 years | 0 | 6 | 3 | 4 | 13 |
| 5–10 years | 2 | 1 | 3 | 4 | 10 |
| 11–20 years | 2 | 0 | 4 | 3 | 9 |
| Over 20 years | 1 | 0 | 5 | 3 | 9 |
| (2) ***Occupation*** | | | | | |
| Building engineering | 0 | 0 | 0 | 11 | 11 |
| Fire safety protection | 9 | 0 | 0 | 5 | 14 |
| Heritage conservation | 0 | 0 | 15 | 0 | 15 |
| Building operations/management | 1 | 12 | 0 | 0 | 13 |
| (3) ***Geographic distribution*** | | | | | |
| Macao SAR | 9 | 12 | 2 | 2 | 25 |
| Hong Kong SAR | 1 | 0 | 0 | 5 | 6 |
| Mainland China | 0 | 0 | 13 | 9 | 22 |
| (4) Building ***type*** *(multiple answer)* | | | | | |
| Residential | 7 | 1 | 2 | 11 | 21 |
| Commercial | 6 | 0 | 3 | 14 | 23 |
| Industrial | 5 | 0 | 1 | 9 | 15 |
| Heritage | 4 | 12 | 15 | 1 | 32 |
| Other (**roads, tunnels, bridges, and parking lots**) | 3 | 0 | 1 | 1 | 5 |
| (5) ***Work experience*** *(multiple answer)* | | | | | |
| Fire safety design | 1 | 0 | 1 | 5 | 7 |
| Fire risk assessment | 2 | 5 | 3 | 7 | 17 |
| On-site emergency response | 5 | 11 | 6 | 8 | 30 |
| Routine fire safety inspection | 5 | 9 | 4 | 10 | 28 |
| Building safety compliance review | 1 | 4 | 8 | 9 | 22 |
| Other (**fire investigation, fire education, and wildfire suppression**) | 5 | 0 | 0 | 1 | 6 |

Note.
[a] FES, HBM, CHC, and BSE refer to the professional groups of fire emergency services, heritage building management, cultural heritage conservation, and building safety engineering, respectively.

Fig. 6; detailed results are provided in Appendix D.1.

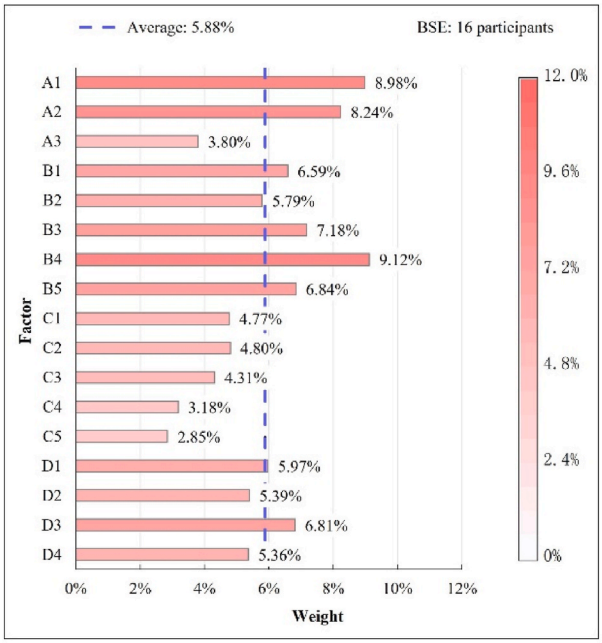
#### 4.2.1. Category-level weight distribution

From a holistic perspective, professional groups generally considered *fire hazards* (A) and *heritage and firefighting equipment* (B) as the top priorities for risk assessment and management. In contrast, the average rankings of *surrounding environment* (C) and *fire safety management* (D) were relatively lower. Professionals from all four groups consistently noted that fire risk assessments should focus more on fire hazards, building structures, and fire protection equipment characteristics, especially before and during a fire, while the role of the surrounding environment and fire safety management may be more related to daily management, emergency response, or cleanup after a fire.

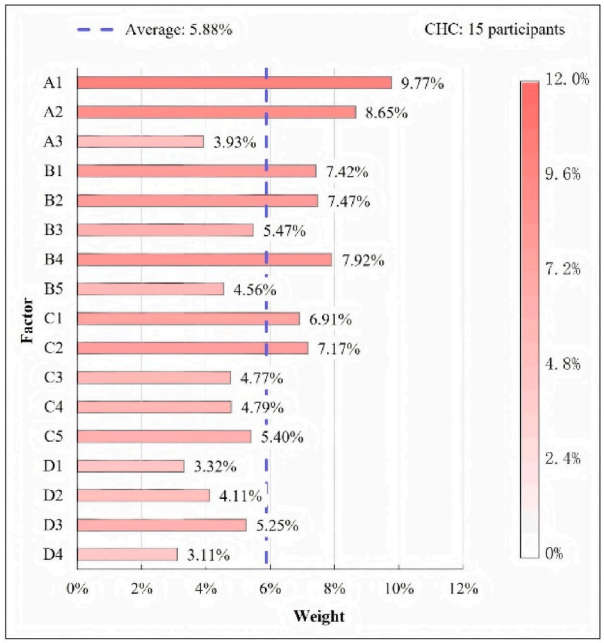
#### 4.2.2. Factor-level weight distribution

In *fire hazards* (A), all four professional groups recognized *ignition sources* (A1) and *combustible and flammable materials* (A2) as the most direct fire hazards and gave them relatively high weights (7.95 %–11.43 %). Specifically, the professionals emphasized that these two factors are the key causes in the formation and spread of fires. *Natural hazards* (A3) received lower and more varied weightings across professional groups than the above two factors. Some professionals noted that despite natural events potentially causing fires, their rarity makes them less prioritized when allocating protection resources. HBM professionals assigned relatively high importance (6.26 %, ranked 8th), probably because their work involves routine building inspections and maintenance that would target seasonal weather changes.

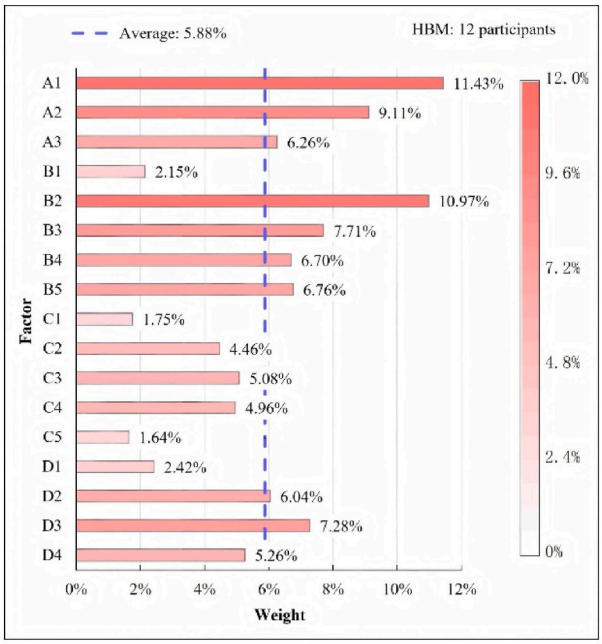
In *heritage and firefighting equipment* (B), the *integrated fire service installations* (B4) was recognized as one of the most important elements by four professional groups, consistently ranking high weights from 6.70 % to 11.47 %. FES members explained that the integrated fire protection facilities and water sources are extremely crucial as they help suppress fires in the first place. For *building inherent features* (B1), the FES, CHC and BSE groups ranked it relatively high (2nd, 5th and 7th, respectively), in contrast to the 15th place ranked by the HBM group (2.15 %). The varying rankings reflected the differences in priorities among the four professional groups. FES valued B1 for fire suppression and rescue deployment, while BSE and CHC highlighted the role of building materials and structures in fire spread and heritage integrity. HBM considered B1 less important, as their duties rarely involve altering inherent building features. By contrast, they believed the *means of egress* (B2) is very important (2nd, 10.97 %) because it is directly related to their responsibilities of assisting occupants and visitors in emergency evacuation.



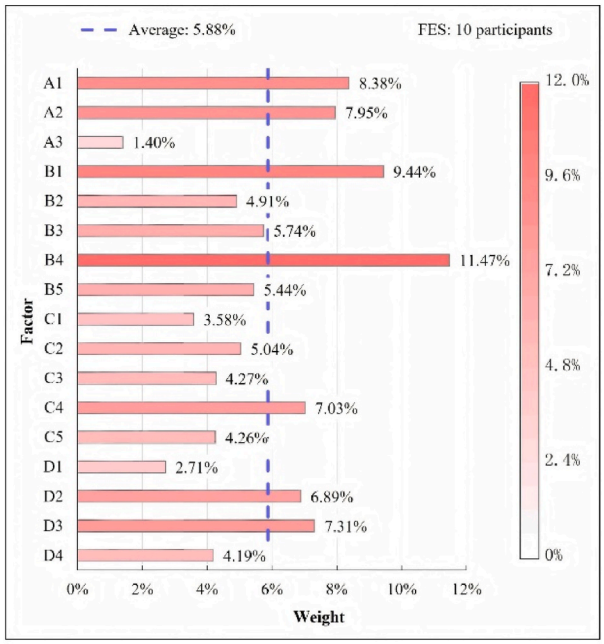

(a) BSE weightings




(b) CHC weightings




(c) HBM weightings




(d) FES weightings

**Fig. 6.** Fire risk factor weightings by the four professional groups.

Regarding *surrounding environment* (C), the average rankings of the factors were largely below average. The professionals had a consensus that they have relatively limited control over the external environment, and these factors had a less important impact on the heritage fire in the early stages of an emergency. Particularly for *surrounding potential dangers* (C5), which was ranked as the least important factor by both the BSE and HBM groups. HBM professionals noted that flammable and explosive hazards (such as gas stations and chemical tank farms) are uncommon in historic districts, and a BSE professional pointed out that the safety distances between external hazards and heritage buildings typically meet regulations. Several disparities were also observed across different professional groups. The factor *external space characteristics* (C1) was ranked intermediately high by the CHC group (7th, 6.91 %) but was deemed less important by the HBM (16th, 1.75 %) and the BSE (13th, 4.77 %) groups. This discrepancy is probably because the daily work of the heritage conservators involves block planning and renovation, with a focus on managing and improving external spaces to reduce the threat of fire to the heritages. In addition, the factor *surrounding water supplies* (C4) was ranked 6th (7.03 %) by the FES group, which is considerably higher than the other groups. This is because the availability and condition of external water supplies, such as fire hydrants, are critical for firefighters to extinguish fires.

For *fire safety management* (D), the weight rankings of most factors ranged from less important to moderately important, while some factors still received relatively high attention. Among them, *routine maintenance* (D3) was ranked relatively high, with a range of 5.25 %–7.31 %. The survey participants had a high consensus on it, where they all believed that daily maintenance is crucial in the entire safety management as it ensures the critical equipment is readily useable in fire emergencies. In addition, *emergency plans* (D2) was rated relatively high by the FES group (7th, 6.89 %), interpreted as "*contingency plans are the key to ensure the fastest and most accurate rescue in a disaster*."

### 4.2.3. Assessment practices and challenges in heritage fire risk management

Regarding the fire risk considerations in current practice, all four professional groups showed a strong tendency to consider fire risks in their daily work, though with varying degrees of emphasis, as shown in Fig. 7. The FES group showed relatively thorough consideration, with a particularly high rate for *fire hazards* (70 %) and *heritage and firefighting equipment* (60 %), reflecting their specialized focus on active fire prevention. The HBM group exhibited a distinct pattern. They placed strong emphasis on *heritage and firefighting equipment* (58.33 % large consideration). Meanwhile, the CHC group showed the highest level of attention to *fire safety management* (53.33 % full consideration) and *fire hazards* (66.67 % full consideration). This reflects a combination of administrative controls and proactive risk mitigation in their work. The BSE group prioritized *fire hazards* (68.75 % full consideration) while having relatively low consideration for *surrounding environment* (18.75 % no consideration).

The survey results further show varying satisfaction levels among the professional groups (as shown in Fig. 8). Among them, HBM showed much higher satisfaction (83.33 %), suggesting their current methods work well for their needs, while CHC was more critical (40 % dissatisfied and 40 % neutral). This may be due to their limited involvement in assessment processes and concerns about insufficient follow-up actions after assessments. FES and BSE remained largely neutral. The result suggests that there is room to improve existing fire risk assessment methods with multi-stakeholder involvements.

When asked about the challenges in existing fire risk assessments, nearly half of the professionals pointed out that insufficient data and inefficient data collection (49 %) as well as inadequate information sharing and collaboration (49 %) are the most prominent

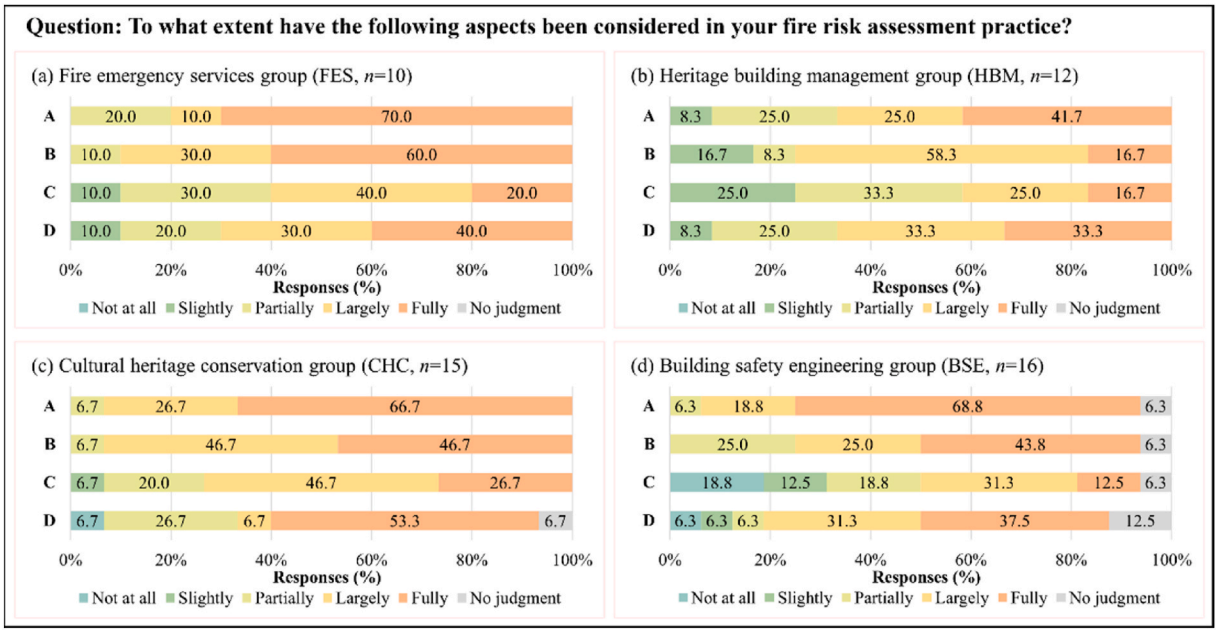



**Fig. 7.** Fire risk considerations in current fire risk assessment practice.

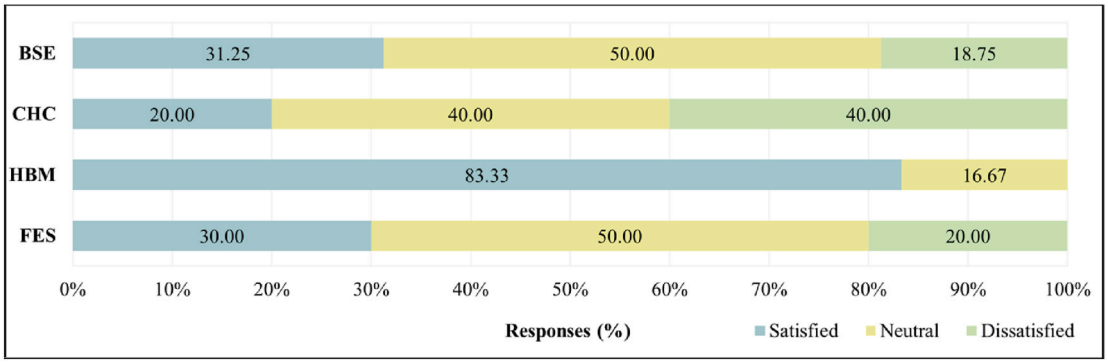



**Fig. 8.** Adequacy of current fire risk assessment practice ($n = 53$).

challenges. The professionals affirmed that the issue of data silos between different parties hinders coordinated monitoring and emergency response for heritage buildings. In addition, over one-third of respondents mentioned the lack of a standardized quantitative assessment method (43 %) and the lack of automated fire risk computation (38 %) as significant challenges. Several professionals added that it is critical for professionals from different backgrounds to conduct a comprehensive assessment of heritage buildings, and they believed the focus on heritage fire safety should shift from reactive response to proactive prevention through digital solutions. However, they reported that the use of automation and robotics technologies (e.g., drones) for heritage building inspections remains limited in current practice.

### *4.3. Professional-centric risk assessment results*

In accordance with the national regulations of fire risk assessments [92], professional-centric assessment involved nine local practitioners across the three selected heritage buildings. Each site engaged three participants: St. Augustine's Church and Lou Kau Mansion each had two heritage building managers and one safety officer; Mandarin's House had two safety officers and one heritage site specialist. All participants are designated fire safety and heritage management personnel employed by the heritage buildings, with four (safety officers) having relevant professional qualifications [93]. Their responses were collected using Q2 and were analyzed using the AHP-based methods introduced in Section 3.3. The professional-centric risk assessment results are presented in Fig. 9.

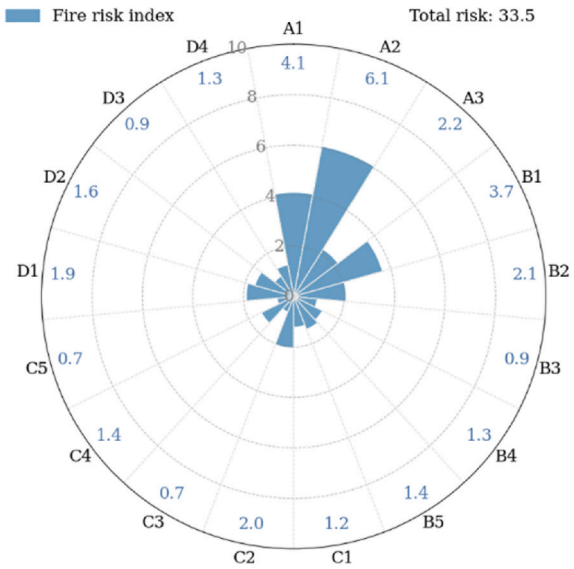
As shown in Fig. 9, the three heritage buildings were all at a moderate-to low-risk level according to the risk level classification, ranging from 33.5 (St. Augustine's Church), 34.3 (Lou Kau Mansion), to 40.1 (Mandarin's House). From risk comparison, *combustible/flammable materials* (A2) were the most significant risk factor for all buildings due to extensive wooden elements in structures and furnishings. *Inherent building features* (B1) ranked within the top four factors for all buildings.

For St. Augustine's Church, *combustible and flammable materials* (A2) was the main risk factor (RI = 6.1) due to extensive wooden elements in benches, worship furniture, and roof structures. Correspondingly, the fire-resistant capabilities of the building interiors remain limited, as reflected in the risk index of *building inherent features* (B1, RI = 3.7). Moreover, the *ignition sources* (A1) was ranked 2nd, which reflects the concerns about the burning candles and the potential outage of old electrical equipment in the church. Notably, since St. Augustine's Church contains naked flames, the risk index of *ignition sources* (4.1) was much higher than the other heritage buildings (1.8). Regarding the *natural disasters* (A3), the survey respondents ranked it 4th as the church is located on a hilltop, which increases the risk of lightning strikes and secondary fires. The hilltop terrain also affected the surrounding transportation, where traffic congestion sometimes happens to hinder emergency response, posing a relatively high risk (2.0) in *road traffic* (C2).

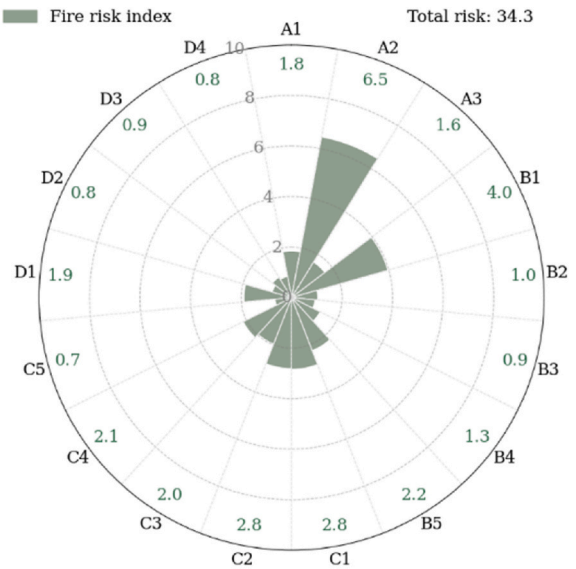
For Lou Kau Mansion, *combustible and flammable materials* (A2) was ranked highest (6.5) due to wooden doors, windows and stairs. *Building inherent features* (B1) was ranked 2nd (4.0) due to poor fire resistance of historical materials and deteriorated wooden beams, floor joists, and roof trusses. *External spatial characteristics* (C1, 3rd) and *road traffic* (C2, 4th) were also ranked high because the heritage site is in a dense tourism area, surrounded by a food street pedestrian zone. Emergency vehicles cannot directly access the building and can only rely on alternative nearby routes.

Regarding Mandarin's House, *combustible and flammable materials* (A2) dominated risks (8.5) due to numerous wooden trusses, columns and stairs. Heavy tourist traffic and rule violations (such as smoking) exacerbate *occupancy status* risks (D1, 3.6). In addition, *external spatial features* (C1, 3.9) was ranked 2nd due to limited fire engine access. The street adjacent to the main entrance fails to meet the minimum width requirement, while the connecting road on the southern side barely meets the requirement, which creates considerable constraints for emergency response.

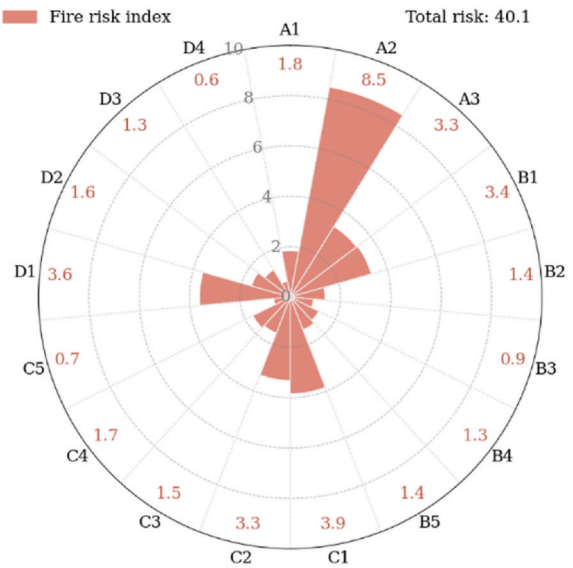
Sensitivity analysis across all perturbation scenarios confirmed the robustness of risk rankings. Mandarin's House maintained the highest-risk position in all scenarios (Spearman $\rho = 1.00$). The findings indicate that moderate variations in risk weights would not significantly affect the overall risk ranking, confirming the reliability of the RATER assessment. Detailed analysis results are presented in Appendix D.3.




(a) St. Augustine's Church




(b) Lou Kau Mansion




(c) Mandarin's House

**Fig. 9.** Fire risk indices of the three heritage sites.

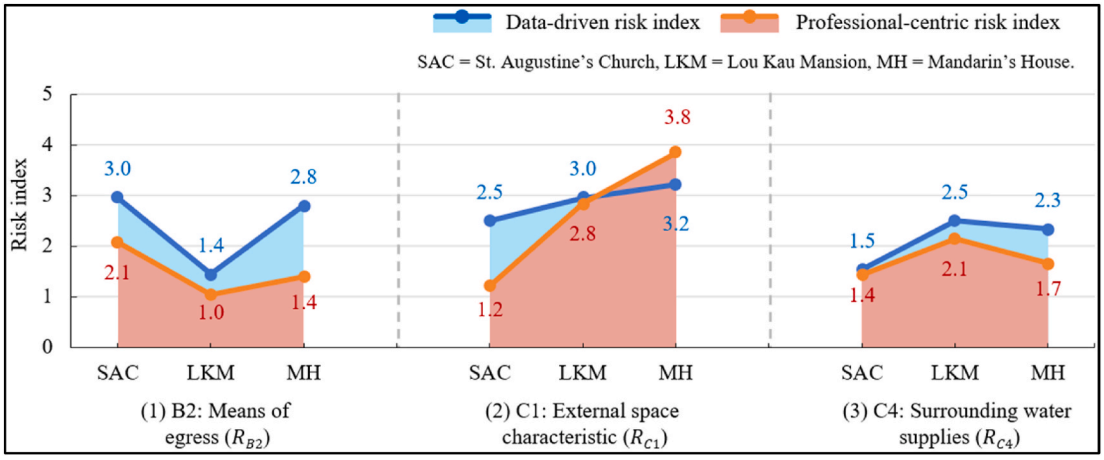



**Fig. 10.** Comparison of risk indices between data-driven and professional-centric assessment.

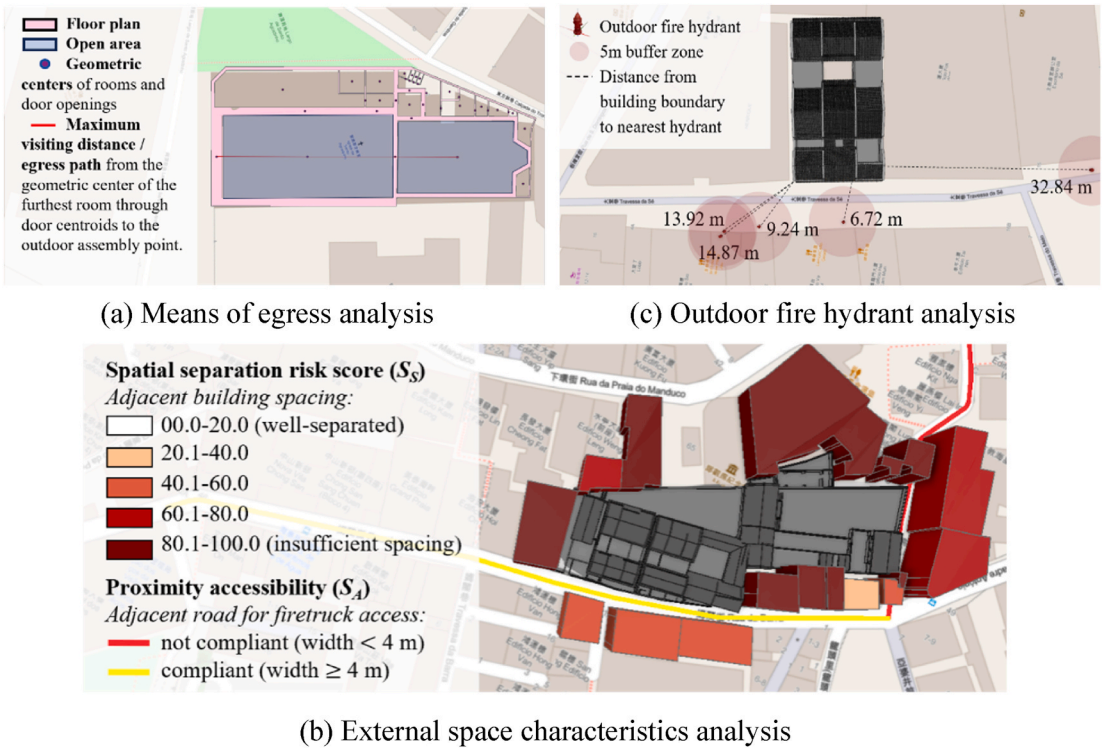



**Fig. 11.** Fire risk assessment visualization using the BIM-GIS integrated system.

### *4.4. BIM-GIS integrated data-driven risk assessment results*

Following the professional-centric assessment, this study leveraged the data in BIM models and GIS maps to conduct an objective risk assessment for cross-checking. The BIM models of the three heritage buildings, as well as the GIS data in OSM and Macao Cartography and Cadaster Bureau [87], were extracted to compute the risk indices using Equations (12)–(21). Consequently, Fig. 10 illustrates the results of both the professional-centric and data-driven assessments. It should be noted that the two assessments cannot be directly compared numerically, as they are based on different measurement logics. Instead, their consistency was examined through comparative trend analysis. Specifically, for each factor, both approaches exhibit consistent ranking patterns across the three heritage buildings, reflecting convergence in relative fire risk levels. This consistency across multiple factors provides mutual validation between the two assessments.

For *means of egress* (B2), the assessment utilized the path length from BIM models to calculate the required evacuation travel time

($T_{req}$). The risk indices for Mandarin's House ($R_{B2} = 2.8$) and St. Augustine's Church ($R_{B2} = 3.0$) were considerably higher than that of Lou Kau Mansion ($R_{B2} = 1.4$), where the three heritages had the lengths of evacuation paths at 41.5 m, 43.9 m, and 21.4 m, respectively. Fig. 11(a) illustrates the path calculation using the proposed system. The $T_{req}$ values of the three buildings were 34.6, 36.6, and 17.9 s, all of which fell below the regulated available safety time ($T_{avl}$), resulting in medium to low risks.

For *external space characteristics* (C1), Mandarin's House demonstrated the highest risk ($R_{C1} = 3.2$), primarily because three of its four adjacent roads fell below the minimum width requirement for fire truck access (4 m), and its average spatial separation distance from surrounding buildings (1.5 m) was much lower than the regulatory minimum (6 m), as shown in Fig. 11(b). Lou Kau Mansion ranked 2nd ($R_{C1} = 3.0$), with two adjacent roads marginally meeting emergency vehicle requirements. St. Augustine's Church exhibited relatively lower risk ($R_{C1} = 2.5$), with one road wider than 5 m that allows fire truck access and provides more surrounding space.

For *surrounding water supplies* (C4), Lou Kau Mansion had the highest risk index ($R_{C4} = 2.5$), followed by Mandarin's House ($R_{C4} = 2.3$) and St. Augustine's Church ($R_{C4} = 1.5$). None of the three buildings fully satisfied the regulations requiring at least two hydrants on street-facing sides and one hydrant on non-street-facing sides, resulting in relatively high risks in the aspect of hydrant quantity ($R_N$). Specifically, St. Augustine's Church had hydrants on three street sides and thus resulting in better coverage. Lou Kau Mansion (Fig. 11(c)) and Mandarin's House had no hydrants distributed on their non-street-facing sides. Regarding the distance between hydrants and heritages, St. Augustine's Church and Lou Kau Mansion largely complied with the requirement to maintain hydrants at least 5 m from building walls, whereas 40 % of Mandarin's House's hydrants were positioned too close to the building ($R_D = 2.0$). These closely positioned hydrants might become inaccessible, owing to exposure to intense heat and falling debris, which in turn makes them less effective for use during fires.

## 5. Discussion

### 5.1. Fire risk factor weighting across divergent professional groups

Fire risk factor weighting analysis across the four professional groups reveals consensus and divergence in risk perceptions. All professional groups prioritize *fire hazards* (A) and *heritage and firefighting equipment* (B), particularly for *ignition sources* (A1), *combustible and flammable materials* (A2) and *integrated fire service installations* (B4), indicating shared recognition of primary fire risk drivers.

On the other hand, the weight distribution patterns also exhibit notable distinctions. FES professionals prioritize *heritage and firefighting equipment* characteristics (B4: 11.47 % and B1: 9.44 %) as well as *fire hazards* (A1: 8.38 % and A2: 7.95 %), aligning with their primary responsibility in firefighting and rescue operations. HBM professionals assign the highest importance to *ignition sources* (A1: 11.43 %) and *means of egress* (B2: 10.97 %), which is consistent with their focus on occupant safety and evacuation during emergencies. The weighted pattern of BSE professionals reflects their technical expertise in integrating building safety systems. Their highest weights are allocated to *integrated fire service installations* (B4: 9.12 %) and *ignition sources* (A1: 8.98 %), while maintaining substantial weights across *combustible and flammable materials* (A2: 8.24 %) and *water supplies* (B5: 6.84 %). CHC professionals show a complementary perspective with domain-specific preservation considerations. They emphasize both material properties and heritage preservation factors, such as *ignition sources* (A1: 9.77 %), *combustible and flammable materials* (A2: 8.65 %), *means of egress* (B2: 7.47 %), and *integrated fire service installations* (B4: 7.92 %). Notably, CHC professionals assign substantially higher weights to *surrounding environment's* factors (C1: 6.91 %, C2: 7.17 %) compared to other professional groups, which reflects their holistic consideration of heritage buildings.

Compared with prior single-discipline studies (e.g., Refs. [36,94]), this approach integrates fire response, heritage management, conservation, and safety engineering viewpoints. The results not only confirm the widely acknowledged importance of ignition sources and combustible materials, but also introduce additional factors such as evacuation routes and external spatial conditions. These aspects, particularly highlighted by the HBM and CHC groups, add new insights to conventional risk perspectives by addressing practical fire safety concerns and spatial environment considerations of heritage sites.

(a) Fire truck access for St. Augustine's Church (b) Unprotected zones around Lou Kau Mansion

**Fig. 12.** Spatial analysis of heritage surroundings based on the BIM-GIS integrated system.

These consensuses and divergences highlight the importance of multi-stakeholder involvement in fire risk assessment, which can further be operationalized both collectively and within specific professional domains. Consensus weights can guide overall policy formulation and balanced resource allocation, while professional-specific results can inform tailored decision-support dashboards and training materials. More importantly, recognizing these divergences also facilitates cross-disciplinary collaboration, enabling each professional to consider factors valued by others. For instance, based on the professional survey, a majority of heritage building managers (58.3 %) give only slight or partial consideration to external surrounding conditions in current practice, whereas most firefighters (60 %) and heritage conservators (73.4 %) consider these external environment data largely or fully necessary, underscoring a considerable disciplinary gap. This gap further suggests that heritage managers, who are often on site and most familiar with day-to-day operations, could strengthen the monitoring of external access paths and water-supply conditions, which firefighters prioritize but cannot routinely oversee in person. Similarly, firefighters can adapt emergency response plans to respect the structural fragility and conservation constraints emphasized by heritage conservators. Heritage conservators may also collaborate with fire safety engineers to adopt non-intrusive detection and suppression solutions suitable for heritage interiors. Such cross-disciplinary awareness integrates complementary expertise and reduces blind spots, helping to remedy the inadequate information sharing and collaboration raised by nearly half of the respondents (49 %).

### 5.2. Synthesis between professional-centric and data-driven risk assessment results

Both professional-centric and data-driven assessments are adopted in this study for fire risk quantification. Specifically, professional-centric assessments enable in-depth discussion with domain professionals to collect first-hand experience and professional knowledge, while data-driven assessments leverage objective data from BIM models and GIS maps to indicate the internal and external risks through standardized formulation. Together, the two approaches complement one another and enable cross-validation of fire risk evaluations of the heritage sites.

The results show strong consistency between the two assessment methods across the three case heritage buildings in Macao. Both methods identify Mandarin's House as the highest-risk site and classify St. Augustine's Church as the lowest risk. Taking a close look into the Mandarin's House, the highest risks identified from the two methods are from *external space characteristics* (C1). During the survey, the heritage building managers identified limitations in emergency operations due to the insufficient roads for fire engine parking based on their observations in previous fire drills. Meanwhile, data-driven analysis provides objective evidence to supplement the professional judgment by measuring the vehicular access and surrounding building separation and comparing them with the regulatory requirements.

Compared to previous studies that only rely on either professional surveys or data-driven analyses, the integrated approach supplements both professional insights and systematic risk quantification. Moreover, the convergence between the two assessments reinforces the validity of the findings, providing more reliable evaluations for decision-making in heritage fire safety management.

### 5.3. BIM-GIS integration for enhanced heritage fire risk management

In addition to the numerical risk results, this study integrates BIM and GIS to develop a 3D system to visualize the heritage fire risks for informed risk mitigation and emergency response. The system incorporates detailed heritage BIM models, surrounding building blocks, road networks, and external hydrant systems to offer intuitive geospatial insights into risk distribution. Taking the *external space characteristics* (C1) as an example, the system explicitly shows that only the northwestern and southeastern roads near St. Augustine's Church satisfy the regulatory requirements for fire truck access, which provides the locational guidance to facilitate emergency preparedness (as shown in Fig. 12(a)). Regarding *surrounding water supplies* (C4), the BIM model indicates building entrance locations, and the GIS map displays the distribution of nearby fire hydrants. Together, the geospatial analysis reveals that both Lou Kou Mansion and Mandarin's House have coverage gaps of fire hydrants on their back street sides, suggesting potential reduced water pressure and delayed fire control caused by long-distance hose relays. Accordingly, supplemental water supplies should be allocated to those unprotected zones, such as the back sides of Lou Kau Mansion illustrated in Fig. 12(b), to mitigate fire risks. By consolidating data on building entrances, hydrant systems, and allowable parking areas, the system allows first responders to make coordinated and spatially informed decisions for firefighting resource allocation. Such analysis is particularly important for space-constrained heritage buildings, yet it remains underexplored in existing research. Therefore, the proposed system has the potential to provide practitioners with actionable geospatial insights to enhance fire risk management and emergency planning, addressing the issues of insufficient data and inefficient data collection (49 %) and lack of automated fire risk computation (38 %) in current practice.

### 5.4. Limitations

During the RATER development and implementation processes, several challenges and limitations existed in weighting calculation, professional-centric and data-driven assessments, as well as case study validation. First, the professional weighting has a relatively narrow geographic and cultural scope. Most professionals came from mainland China, Hong Kong SAR, and Macao SAR, where their judgment on risk weighting may be affected by local safety regulations, management practices, and heritage characteristics. Consequently, the current weighting results may not fully represent international perspectives and should be interpreted with caution. Nevertheless, the RATER model is methodologically transferable as its hierarchical structure and factor definitions are generic and adaptable. Therefore, when applied to other countries or regions, factor weights may be refined through local expert consultation to better reflect region-specific heritage characteristics and fire safety conditions.

Second, in the professional-centric assessments, the case study had a small professional sample for each site. This is, in fact, a common challenge in prior studies and in practice, as only a limited number of field professionals worked directly with the buildings, with each often undertaking multiple roles due to the human resource constraints. Although these professionals possess in-depth knowledge and first-hand experience, their responses may be subject to personal biases, which can affect the representativeness of the findings. To mitigate this influence, cross-validation with the data-driven assessment provides an additional source of measurement to reinforce the reliability of the results.

Third, the data-driven assessment heavily depends on data quality and availability. In this study, factors B2, C1, and C4 are selected since the data of escape route lengths, distances between buildings, and outdoor hydrant locations could be easily obtained from BIM models and public GIS databases. However, other factors are more difficult to assess, as they involve fine-grained building elements such as integrated fire systems that are often omitted from BIM models. Accurate evaluation of these elements requires either highly detailed as-built modeling or manual field surveys for data collection. Similarly, GIS data are valuable for representing surrounding environments, but they sometimes may not reflect the up-to-date situations. In the pilot experiment, publicly accessible data from OSM were leveraged to determine the height of heritage buildings, but the results turned out to be misleading due to inaccuracies in the source data. To address the inaccuracies, the OSM data were cross-checked against the official cadastral and urban planning datasets released by the Macao Cartography and Cadaster Bureau and other authoritative sources to correct the elevation data. These data quality and availability bottlenecks are critical challenges for adopting the BIM-GIS system at other heritage sites worldwide. While the data extraction, analysis, and visualization techniques are methodologically generic, successful application depends on local dataset readiness. Therefore, future work should explore advanced data capture and cleaning methods to enhance data quality and support cross-city implementation of the BIM-GIS system.

Last, the scope of the case studies and the mode of collaboration should be further expanded. Although this study has attempted to include heritages of both Western religious buildings and Chinese courtyard compounds, many other types remain underexplored. In particular, high-risk industrial heritages, large religious complexes (e.g., monasteries or temple ensembles), and open memorial sites in rural areas may exhibit different spatial configurations, fire dynamics, and safety requirements that need recalibration of factor weights and analytical parameters. Hence, a systematic investigation with more heritage buildings across different regions is necessary to improve the generalizability of the assessment methods. Moreover, in the present study, interdisciplinary collaboration was primarily reflected in the fire risk weighting process. Future work should extend such collaboration to the system design stage, where building managers, firefighters, heritage conservators, and fire safety engineers can jointly participate in developing tailored functions and decision-support tools that better support diverse use case scenarios.

In response to the above limitations, several improvements should be made to future work. First, the research should include numerous professionals from more countries and different professional backgrounds. This will improve the applicability of the weighted RATER across different heritage buildings. Second, this study should be expanded to include more types of heritage buildings to comprehensively assess the applicability of the assessment model. Third, digital tools for fire risk modeling require improvement. As the survey respondents pointed out, real-time sensor data, reality capture using drone scanning or LiDAR measurements, data sharing between agencies, and advanced data processing could be added to enhance the data availability, quality, and analytical efficiency. For instance, panoramic capture and reconstruction can enhance the level of detail of BIM models and GIS maps for data enrichment [95]. This digital integration can ultimately support more intelligent and reliable fire risk assessments for heritage buildings as well as facilitate the adoption of the BIM-GIS system in other cities.

## 6. Conclusions

This study proposes and develops a novel risk assessment model for historic centers called RATER. Unlike existing frameworks such as FRAME and ARICA, RATER unifies professional-centric judgments on fire risk importance with data-driven BIM-GIS spatial analysis. First, it incorporates fire hazards, heritage and firefighting equipment, surrounding environment, and fire safety management, covering 17 risk factors in total. Through AHP-based surveys with 53 professionals from firefighters, heritage managers, heritage conservators, and fire safety engineers, the relative weights of these factors are determined to reveal their importance in heritage fire safety. Specifically, the results disclose that *ignition sources* (A1), *combustible and flammable materials* (A2), and *integrated fire service installations* (B4) are the top three factors rated by the different professional groups. Moreover, the results elucidate the distinctive weighting patterns across the professional groups, which in turn highlights the necessity of interdisciplinary perspectives and collaborative teamwork to facilitate multi-faceted heritage fire assessment.

On the basis of the weighted RATER, this study further develops approaches for both professional-centric and data-driven assessments. The assessments are applied to three real-world heritage sites with different architectural styles. Specifically, professional-

centric assessments gather first-hand, detailed insights from domain professionals, while data-driven assessments adopt standardized regulation-based formulations with data from BIM models and GIS systems to enable automated fire risk analysis. The case study results reveal consistent risk trends across the heritage sites, demonstrating that the RATER could effectively quantify fire risks with sufficient interpretative details. Furthermore, the integration of the two assessment approaches provides two independent sources of measurement for cross-validation to reinforce the reliability of the findings. The professional-centric approach is easy to implement but relies heavily on manual inputs and may carry subjective biases. The data-driven approach, while presently limited by data availability and accuracy, has the potential to offer more standardized, scalable, and replicable analyses. Using these approaches in a complementary manner enables both practitioner judgements and computed evidence to support more balanced fire risk evaluation.

Finally, to help better interpret the fire risks of heritage buildings, this study establishes a BIM-GIS integrated system to provide analytical insights into building interior characteristics, evacuation route analysis, external spatial conditions, and fire hydrant coverage. For instance, the system can identify areas with insufficient hydrant coverage (such as the backstreet sides of Lou Kau Mansion and Mandarin's House in the case studies), which offers actionable information for policymakers to optimize water resource allocation and for firefighters to make informed fire suppression strategies. In addition, the system visualizes emergency vehicle access routes that require coordination of heritage managers and fire brigades to maintain unobstructed methods for emergency responses. Furthermore, the BIM-GIS system presents the physical adjacency between heritage buildings and surrounding structures as well as the internal spatial structures, which assist heritage conservators and safety engineers in developing tailored measures to reduce the risk of fire propagation.

Theoretically, the RATER provides a methodological framework for multi-factor, multi-stakeholder fire risk assessment. Practically, the BIM-GIS integrated system forms a technical solution to support data-driven risk management by linking multi-source spatial data, expert-derived weights, and automated computation into a unified and structured assessment framework. This integration also opens opportunities for longitudinal monitoring of fire risk, enabling the tracking of risk changes over time as conservation and urban conditions evolve. Future work should adapt the RATER to a broader range of heritage sites and regions to further validate its applicability. Furthermore, advanced technologies like reality capture and environmental sensing should be integrated into the data-driven assessment and prototyping to facilitate the processes of data collection, analysis, and visualization for intelligent heritage fire risk diagnosis and management.

## CRediT authorship contribution statement

**Yutong Qiao:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chi Chiu Lam:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Honghao Zhong:** Writing – review & editing, Formal analysis, Data curation. **Mi Pan:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis. **Tianhang Zhang:** Writing – review & editing, Investigation, Data curation. **Mun On Wong:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mun On Wong reports financial support was provided by The Science and Technology Development Fund, Macao S.A.R. Chi Chiu Lam reports financial support was provided by The Science and Technology Development Fund, Macao S.A.R. Yutong Qiao reports financial support was provided by The Science and Technology Development Fund, Macao S.A.R. Mi Pan reports financial support was provided by The Science and Technology Development Fund, Macao S.A.R. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Methodological framework and computational procedures of the risk assessment

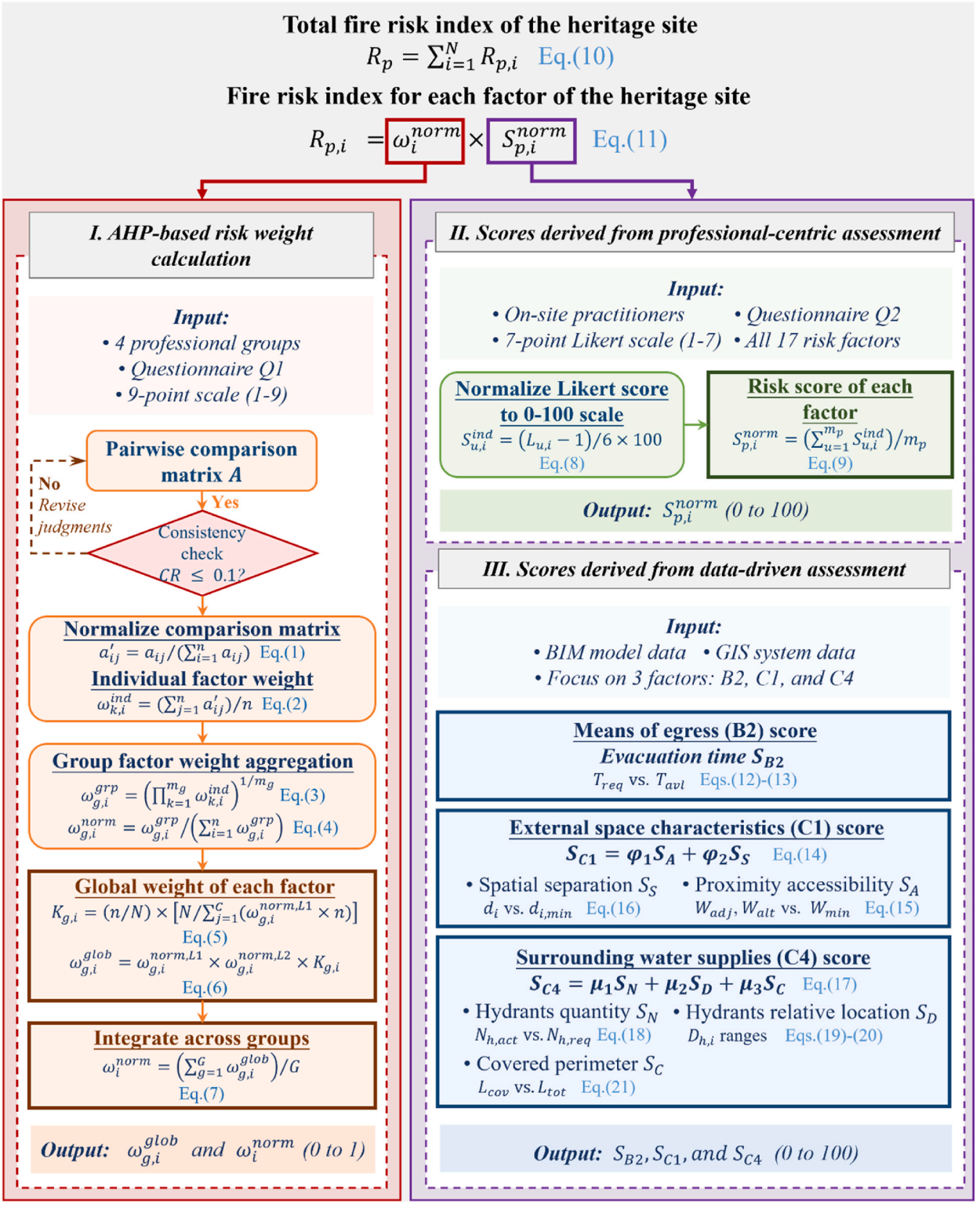


**Fig. A.1.** Methodological framework and computational procedures of the risk assessment.

## Appendix B. Questionnaire Q1–Pairwise comparison sheet for risk factors

This section presents pairwise comparisons of heritage building fire risk factors. Please compare their relative importance.

### *B.1 Category A: Fire hazards*

**Fire hazards** (Category A) refer to factors that significantly increase the likelihood or severity of a fire, considering direct threats such as *ignition sources* (A1), indirect threats like *combustible and flammable materials* (A2), and incidental threats such as *natural hazards* (A3).

**Table B.1**
Pairwise comparison sheet (1)

| **1. Hierarchical ranking**<br>Please rank the factors in Category A in a descending order based on their relative importance (**1: the most importance, 3: the least importance**) | **Rank #** |
|---|---|
| **A1:** Ignition sources | _____ |
| **A2:** Combustible and flammable materials | _____ |
| **A3:** Natural hazards | _____ |

**2. Pairwise comparison** – Please compare the importance ratios of different factor pairs

| | **The left item is more important than the right item** | | | | | | | | **Equal** | **The right item is more important than the left item** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ranked item** | Absolute | ← | High | ← | Essential | ← | Minor | ← | | → | Minor | → | Essential | → | High | → | Absolute | **Ranked item** |
| | 9:1 | 8:1 | 7:1 | 6:1 | 5:1 | 4:1 | 3:1 | 2:1 | 1:1 | 1:2 | 1:3 | 1:4 | 1:5 | 1:6 | 1:7 | 1:8 | 1:9 | |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#2 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |

### *B.2 Category B: Heritage and firefighting equipment*

**Heritage and firefighting equipment** (Category B) encompass the features of buildings and firefighting services that affect fire propagation, emergency evacuation, and fire suppression. Correspondingly, *building inherent features* (B1), *means of egress* (B2), *portable firefighting devices* (B3), *integrated fire service installations* (B4), and *water supplies* (B5) are incorporated in this category.

**Table B.2**
Pairwise comparison sheet (2)

| **1. Hierarchical ranking**<br>Please rank the factors in Category B in a descending order based on their relative importance (**1: the most importance, 5: the least importance**) | **Rank #** |
|---|---|
| **B1:** Building inherent features | _____ |
| **B2:** Means of egress | _____ |
| **B3:** Portable firefighting devices | _____ |
| **B4:** Integrated fire service installations | _____ |
| **B5:** Water supplies | _____ |

**2. Pairwise comparison** – Please compare the importance ratios of different factor pairs

| | **The left item is more important than the right item** | | | | | | | | **Equal** | **The right item is more important than the left item** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ranked item** | Absolute | ← | High | ← | Essential | ← | Minor | ← | | → | Minor | → | Essential | → | High | → | Absolute | **Ranked item** |
| | 9:1 | 8:1 | 7:1 | 6:1 | 5:1 | 4:1 | 3:1 | 2:1 | 1:1 | 1:2 | 1:3 | 1:4 | 1:5 | 1:6 | 1:7 | 1:8 | 1:9 | |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#2 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#4 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |

### *B.3 Category C: Surrounding environment*

**Surrounding environment** (Category C) relates to external conditions that influence the severity and response actions for fires in heritage buildings and historic centers, such as *external space characteristic* (C1), *road traffic* (C2), *firefighting resources* (C3), *surrounding water supplies* (C4), and *surrounding potential dangers* (C5).

**Table B.3**
Pairwise comparison sheet (3)

| **1. Hierarchical ranking**<br>Please rank the factors in Category C in a descending order based on their relative importance (**1: the most importance, 5: the least importance**) | **Rank #** |
|---|---|
| **C1:** External space characteristic | |
| **C2:** Road traffic | |
| **C3:** Firefighting resources | |
| **C4:** Surrounding water supplies | |
| **C5:** Surrounding potential dangers | |

**2. Pairwise comparison** – Please compare the importance ratios of different factor pairs

| | **The left item is more important than the right item** | | | | | | | | **Equal** | **The right item is more important than the left item** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ranked item** | Absolute | ← | High | ← | Essential | ← | Minor | ← | | → | Minor | → | Essential | → | High | → | Absolute | **Ranked item** |
| | 9:1 | 8:1 | 7:1 | 6:1 | 5:1 | 4:1 | 3:1 | 2:1 | 1:1 | 1:2 | 1:3 | 1:4 | 1:5 | 1:6 | 1:7 | 1:8 | 1:9 | |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#2 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |
| **#4 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#5 item** |

### *B.4 Category D: Fire safety management*

**Fire safety management** (Category D) incorporates the plans and practices among stakeholders in order to reduce the risks and impacts of fires in heritage buildings, such as *occupancy status* (D1), *emergency plans* (D2), *routine maintenance* (D3), and *multi-stakeholder training* (D4).

**Table B.4**
Pairwise comparison sheet (4)

| **1. Hierarchical ranking**<br>Please rank the factors in Category D in a descending order based on their relative importance (**1: the most importance, 4: the least importance**) | **Rank #** |
|---|---|
| **D1:** Occupancy status | |
| **D2:** Emergency plans | |
| **D3:** Routine maintenance | |
| **D4:** Multi-stakeholder training | |

**2. Pairwise comparison** – Please compare the importance ratios of different factor pairs

| | **The left item is more important than the right item** | | | | | | | | **Equal** | **The right item is more important than the left item** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ranked item** | Absolute | ← | High | ← | Essential | ← | Minor | ← | | → | Minor | → | Essential | → | High | → | Absolute | **Ranked item** |
| | 9:1 | 8:1 | 7:1 | 6:1 | 5:1 | 4:1 | 3:1 | 2:1 | 1:1 | 1:2 | 1:3 | 1:4 | 1:5 | 1:6 | 1:7 | 1:8 | 1:9 | |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#2 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |

*B.5 Four categories*

Based on the source and nature of risks, **fire risk factors** can be categorized into four groups.

*Fire hazards* (A): ignition sources, combustible and flammable materials, natural hazards;

*Heritage and firefighting equipment* (B): building inherent features, means of egress, portable firefighting devices, integrated fire service installations, water supplies;

*Surrounding environment* (C): external space characteristic, road traffic, firefighting resources, surrounding water supplies, surrounding potential dangers;

*Fire safety management* (D): occupancy status, emergency plans, routine maintenance, multi-stakeholder training.

**Table B.5**
Pairwise comparison sheet (5)

| **1. Hierarchical ranking**<br>Please rank the categories in a descending order based on their relative importance (**1: the most importance, 4: the least importance**) | **Rank #** |
|---|---|
| **A:** Fire hazards | _____ |
| **B:** Heritage and firefighting equipment | _____ |
| **C:** Surrounding environment | _____ |
| **D:** Fire safety management | _____ |

**2. Pairwise comparison** – Please compare the importance ratios of different factor pairs

| | **The left item is more important than the right item** | | | | | | | | **Equal** | **The right item is more important than the left item** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Ranked item** | Absolute | ← | High | ← | Essential | ← | Minor | ← | | → | Minor | → | Essential | → | High | → | Absolute | **Ranked item** |
| | 9:1 | 8:1 | 7:1 | 6:1 | 5:1 | 4:1 | 3:1 | 2:1 | 1:1 | 1:2 | 1:3 | 1:4 | 1:5 | 1:6 | 1:7 | 1:8 | 1:9 | |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#2 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#1 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#3 item** |
| **#2 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |
| **#3 item** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | **#4 item** |

## Appendix C. Questionnaire Q2–Expert evaluation of heritage building fire risk factors

This section asks you to assess the degree of fire risk posed by different factors in a specific heritage building on a **seven-point Likert scale**, where **1 = very low risk** and **7 = very high risk**. For each factor, a short definition is provided followed by guidelines describing what each level (1–7) represents. Please base your rating on the building under consideration.

*C.1 Category A: Fire hazards*

**Fire hazards** (Category A) refer to factors that significantly increase the likelihood or severity of a fire, considering direct threats such as *ignition sources* (A1), indirect threats like *combustible and flammable materials* (A2), and incidental threats such as *natural hazards* (A3).

**Table C.1**
Professional assessment-based risk scores for Category A

| **What is the current fire risk status of the following factors?** | **1 - Very low** | **2 - Low** | **3 - Slightly low** | **4 - Moderate** | **5 - Slightly high** | **6 - High** | **7 - Very high** |
|---|---|---|---|---|---|---|---|
| **Ignition sources** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| **Combustible and flammable materials** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| **Natural hazards** | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

**Ignition sources**: Agents that produce sparks, flames, or heat to initiate combustion, including naked lights, electrical faults, and hot surfaces.

**Combustible and flammable materials**: Combustible and flammable materials are items that can easily ignite or burn to sustain combustion when exposed to an ignition source, such as furniture and furnishings, wood, textiles, plastics, and gas systems.

**Natural hazards**: Natural events that increase the likelihood or severity of fires in heritage buildings, such as lightning strikes, earthquake-induced ignitions, and extreme climatic conditions (e.g., typhoons, droughts, and heatwaves).

*C.2 Category B: Heritage and firefighting equipment*

**Heritage and firefighting equipment** (Category B) encompass the features of buildings and firefighting services that affect fire propagation, emergency evacuation, and fire suppression. Correspondingly, *building inherent features* (B1), *means of egress* (B2), *portable firefighting devices* (B3), *integrated fire service installations* (B4), and *water supplies* (B5) are incorporated in this category.

**Table C.2**
Professional assessment-based risk scores for Category B

| What is the current fire risk status of the following factors? | 1 - Very low | 2 - Low | 3 - Slightly low | 4 - Moderate | 5 - Slightly high | 6 - High | 7 - Very high |
|---|---|---|---|---|---|---|---|
| Building inherent features | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Means of egress | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Portable firefighting devices | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Integrated fire service installations | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Water supplies | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

**Building inherent features**: Building attributes that affect fire spread and propagation, such as fire resistance, building safety layouts (e.g., fire compartmentation and separation), and functional areas (e.g., hazardous storage areas).

**Means of egress**: Means of egress are the pathways and exits designed to assist occupants and firefighters in escape and rescue operations.

**Portable firefighting devices**: Apparatuses that can be easily carried to combat fires in heritage buildings, e.g., fire extinguishers and blankets.

**Integrated fire service installations**: Fixed systems installed within heritage buildings that assist occupants and firefighters in fire response, including alarms, sensors, sprinkler systems, gas flooding systems, and emergency electricity and lighting systems.

**Water supplies**: Sources and apparatuses designed to supply water with sufficient pressures and flow rates, e.g., pipe systems, tanks and pumps.

*C.3 Category C: Surrounding environments*

**Surrounding environment** (Category C) relates to external conditions that influence the severity and response actions for fires in heritage buildings and historic centers, such as *external space characteristic* (C1), *road traffic* (C2), *firefighting resources* (C3), *surrounding water supplies* (C4), and *surrounding potential dangers* (C5).

**Table C.3**
Professional assessment-based risk scores for Category C

| What is the current fire risk status of the following factors? | 1 - Very low | 2 - Low | 3 - Slightly low | 4 - Moderate | 5 - Slightly high | 6 - High | 7 - Very high |
|---|---|---|---|---|---|---|---|
| External space characteristics | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Road traffic | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Firefighting resources | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Surrounding water supplies | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Surrounding potential dangers | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

**External space characteristics**: Spatial conditions that determine the exposure and isolation of heritage buildings and affect the spread and severity of fires, such as fire separation distance, frontage, geographical elevations, and adjacent structures.

**Road traffic**: Transportation situations that affect the accessibility and mobility of first responders and fire engines toward the affected heritage buildings, covering static road information and dynamic traffic flow information.

**Firefighting resources**: Facilities, engines, and human resources allocated for fire responses in heritage buildings.

**Surrounding water supplies**: Sources of water available in the vicinity of heritage buildings that can be used to combat fires, such as fire hydrants and natural water bodies.

**Surrounding potential dangers**: Adverse elements in the vicinity of heritage buildings that may affect the severity of fires, such as gas stations, vegetation, and power lines.

*C.4 Category D: Fire safety management*

**Fire safety management** (Category D) incorporates the plans and practices among stakeholders in order to reduce the risks and impacts of fires in heritage buildings, such as *occupancy status* (D1), *emergency plans* (D2), *routine maintenance* (D3), and *multi-stakeholder training* (D4).

**Table C.4**
Professional assessment-based risk scores for Category D

| What is the current fire risk status of the following factors? | 1 - Very low | 2 - Low | 3 - Slightly low | 4 - Moderate | 5 - Slightly high | 6 - High | 7 - Very high |
|---|---|---|---|---|---|---|---|
| Occupancy status | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Emergency plans | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Routine maintenance | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |
| Multi-stakeholder training | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

**Occupancy status**: Occupancy conditions where the heritage building is generally occupied and used, including regular (e.g., tourism), occasional (e.g., cultural events), and temporary activities (e.g., renovation work).

**Emergency plans**: Protocols that regulate the roles, responsibilities, procedures, and resources of different emergency participants involved in preventing, preparing, and responding to fire incidents in heritages.

**Routine maintenance**: Activities of inspecting, testing, repairing, and replacing heritage structures and firefighting equipment to ensure functionality against fire incidents in heritage buildings.

**Multi-stakeholder training**: Joint efforts that involve various parties in training activities to practice fire responses in heritage buildings.

## Appendix D. Factor weight analysis, reliability assessment, and sensitivity testing

*D.1 Factor risk weights comparison results using Questionnaire 1*

**Table D.1**
The normalized factor weights (%) assigned by participants in the BSE group (with conditional formatting)

| | Factor risk weight | | | | | | | | | | | | | | | | | CR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A1 | A2 | A1 | A2 | A1 | A2 | A1 | A2 | A1 | A2 | A1 | A2 | A1 | A2 | A1 | D3 | D4 | |
| **Group (GM)** | **9.0** | **8.2** | **3.8** | **6.6** | **5.8** | **7.2** | **9.1** | **6.8** | **4.8** | **4.8** | **4.3** | **3.2** | **2.9** | **6.0** | **5.4** | **6.8** | **5.4** | - |
| **BSE 01** | 8.7 | 16.7 | 1.8 | 12.1 | 9.3 | 4.1 | 7.5 | 1.4 | 4.1 | 3.4 | 1.8 | 0.6 | 5.4 | 8.4 | 2.9 | 5.4 | 6.4 | 0.017 |
| **BSE 02** | 1.1 | 9.5 | 8.7 | 12.6 | 7.1 | 7.1 | 8.4 | 7.1 | 6.9 | 5.3 | 3.9 | 3.9 | 6.9 | 1.5 | 3.2 | 3.6 | 3.2 | 0.017 |
| **BSE 03** | 6.6 | 7.6 | 2.9 | 9.3 | 7.0 | 8.0 | 8.0 | 8.0 | 6.8 | 3.4 | 3.4 | 3.4 | 6.8 | 5.3 | 2.3 | 6.9 | 4.4 | 0.022 |
| **BSE 04** | 9.3 | 3.6 | 8.1 | 0.8 | 2.5 | 4.6 | 5.7 | 7.5 | 3.6 | 10.5 | 6.5 | 8.0 | 1.2 | 4.9 | 8.1 | 6.9 | 8.1 | 0.022 |
| **BSE 05** | 2.4 | 5.4 | 6.2 | 11.9 | 6.7 | 6.7 | 7.9 | 6.7 | 4.6 | 3.9 | 3.9 | 3.9 | 7.0 | 2.8 | 6.3 | 8.3 | 5.3 | 0.022 |
| **BSE 06** | 12.4 | 12.4 | 2.5 | 6.9 | 6.9 | 6.9 | 6.9 | 6.9 | 4.3 | 4.3 | 3.6 | 2.6 | 0.5 | 7.0 | 2.6 | 7.0 | 6.4 | 0.017 |
| **BSE 07** | 9.9 | 11.4 | 4.3 | 12.0 | 12.0 | 7.2 | 10.1 | 1.4 | 4.4 | 2.6 | 2.6 | 0.5 | 3.2 | 1.0 | 6.1 | 5.6 | 5.6 | 0.019 |
| **BSE 08** | 9.5 | 5.2 | 2.9 | 6.9 | 2.7 | 6.4 | 6.9 | 6.4 | 5.9 | 5.9 | 5.9 | 5.9 | 5.9 | 4.9 | 5.8 | 7.0 | 5.8 | 0.022 |
| **BSE 09** | 7.3 | 4.6 | 5.8 | 4.5 | 5.8 | 5.8 | 6.7 | 6.7 | 5.1 | 6.8 | 5.8 | 5.8 | 5.8 | 5.9 | 5.9 | 5.9 | 5.9 | 0.056 |
| **BSE 10** | 9.5 | 5.2 | 2.9 | 5.9 | 5.9 | 5.9 | 5.9 | 5.9 | 6.8 | 6.8 | 6.8 | 6.8 | 2.3 | 17.6 | 2.0 | 2.0 | 2.0 | 0.010 |
| **BSE 11** | 10.9 | 5.6 | 1.2 | 1.2 | 3.5 | 7.9 | 10.4 | 6.4 | 2.7 | 6.3 | 7.8 | 7.8 | 4.8 | 6.6 | 6.6 | 7.3 | 3.0 | 0.013 |
| **BSE 12** | 6.1 | 6.8 | 1.1 | 7.4 | 9.7 | 4.1 | 12.0 | 12.0 | 1.1 | 1.2 | 1.2 | 1.2 | 0.1 | 13.1 | 5.7 | 15.5 | 1.8 | 0.013 |
| **BSE 13** | 8.7 | 1.8 | 16.7 | 12.1 | 1.4 | 4.1 | 9.3 | 7.5 | 5.1 | 3.7 | 3.0 | 3.0 | 0.6 | 15.8 | 3.0 | 1.8 | 2.3 | 0.017 |
| **BSE 14** | 7.1 | 13.7 | 1.5 | 7.4 | 7.4 | 7.4 | 7.4 | 7.4 | 3.1 | 3.1 | 3.1 | 3.1 | 1.5 | 6.6 | 6.6 | 5.6 | 8.0 | 0.022 |
| **BSE 15** | 14.4 | 4.8 | 1.6 | 1.3 | 1.3 | 11.0 | 11.0 | 10.0 | 5.0 | 5.0 | 5.0 | 0.7 | 5.3 | 2.4 | 7.1 | 7.1 | 7.1 | 0.022 |
| **BSE 16** | 13.3 | 12.3 | 1.6 | 6.9 | 6.9 | 6.9 | 6.9 | 6.9 | 3.1 | 3.1 | 3.1 | 3.1 | 3.1 | 2.9 | 6.4 | 8.4 | 5.4 | 0.017 |
| **Arithmetic mean** | 8.6 | 7.9 | 4.4 | 7.5 | 6.0 | 6.5 | 8.2 | 6.8 | 4.5 | 4.7 | 4.2 | 3.8 | 3.8 | 6.7 | 5.0 | 6.5 | 5.0 | - |
| **Maximum** | 14.4 | 16.7 | 16.7 | 12.6 | 12.0 | 11.0 | 12.0 | 12.0 | 6.9 | 10.5 | 7.8 | 8.0 | 7.0 | 17.6 | 8.1 | 15.5 | 8.1 | - |
| **Minimum** | 1.1 | 1.8 | 1.1 | 0.8 | 1.3 | 4.1 | 5.7 | 1.4 | 1.1 | 1.2 | 1.2 | 0.5 | 0.1 | 1.0 | 2.0 | 1.8 | 1.8 | - |

***Note.*** (1) Weights were derived by combining category-level and factor-level judgments and adjusted to remove bias from unequal category sizes. (2) All values were normalized to a total of 100 percent and are expressed as percentages (one decimal place). (3) A unified green–yellow gradient color scale (midpoint = 5.88%) visualizes relative weight importance, with greener tones indicating higher values.

**Table D.2**
The normalized factor weights (%) assigned by participants in the CHC group (with conditional formatting)

| | Factor risk weight | | | | | | | | | | | | | | | | | CR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A1 | A2 | A3 | B1 | B2 | B3 | B4 | B5 | C1 | C2 | C3 | C4 | C5 | D1 | D2 | D3 | D4 | |
| **Group (GM)** | **9.8** | **8.6** | **3.9** | **7.4** | **7.5** | **5.5** | **7.9** | **4.6** | **6.9** | **7.2** | **4.8** | **4.8** | **5.4** | **3.3** | **4.1** | **5.3** | **3.1** | - |
| **CHC 01** | 7.8 | 6.8 | 3.0 | 1.0 | 4.8 | 9.4 | 9.4 | 4.8 | 1.0 | 4.8 | 4.8 | 9.4 | 9.4 | 5.5 | 2.9 | 8.6 | 6.5 | 0.019 |
| **CHC 02** | 4.6 | 9.1 | 9.1 | 9.8 | 7.8 | 3.6 | 7.8 | 6.8 | 12.8 | 12.8 | 1.4 | 4.0 | 6.9 | 0.2 | 1.3 | 1.5 | 0.6 | 0.012 |
| **CHC 03** | 4.4 | 8.9 | 8.9 | 9.9 | 7.8 | 3.6 | 7.8 | 6.8 | 12.5 | 12.5 | 2.3 | 4.1 | 6.9 | 0.2 | 1.4 | 1.6 | 0.6 | 0.012 |
| **CHC 04** | 7.1 | 7.1 | 3.5 | 7.1 | 9.6 | 3.2 | 3.2 | 6.2 | 5.8 | 5.8 | 5.1 | 5.8 | 6.8 | 7.0 | 5.8 | 5.8 | 4.9 | 0.022 |
| **CHC 05** | 11.9 | 6.1 | 1.3 | 13.9 | 10.3 | 4.7 | 8.9 | 4.7 | 8.4 | 5.0 | 5.0 | 5.9 | 2.6 | 4.2 | 2.7 | 3.2 | 1.4 | 0.017 |
| **CHC 06** | 11.0 | 6.1 | 3.3 | 6.8 | 6.8 | 6.8 | 6.8 | 6.8 | 7.8 | 6.3 | 1.2 | 3.5 | 10.2 | 0.7 | 6.3 | 6.3 | 3.2 | 0.022 |
| **CHC 07** | 22.0 | 22.0 | 22.0 | 0.5 | 2.7 | 1.5 | 3.3 | 4.3 | 3.3 | 1.5 | 2.7 | 0.5 | 4.3 | 3.5 | 0.5 | 4.2 | 1.5 | 0.009 |
| **CHC 08** | 16.3 | 8.4 | 1.7 | 11.2 | 1.2 | 6.5 | 8.1 | 6.5 | 9.9 | 7.6 | 6.1 | 3.3 | 1.1 | 6.1 | 3.3 | 1.9 | 0.7 | 0.017 |
| **CHC 09** | 7.6 | 7.6 | 2.5 | 6.6 | 6.6 | 6.6 | 5.8 | 3.9 | 5.9 | 5.9 | 5.9 | 5.9 | 5.9 | 4.1 | 6.8 | 5.8 | 6.8 | 0.022 |
| **CHC 10** | 2.7 | 5.3 | 0.6 | 12.6 | 6.8 | 3.8 | 6.8 | 1.3 | 10.1 | 8.5 | 3.9 | 3.9 | 8.5 | 8.7 | 5.1 | 5.1 | 6.2 | 0.022 |
| **CHC 11** | 7.3 | 4.6 | 5.8 | 5.8 | 8.7 | 4.9 | 4.9 | 4.9 | 5.0 | 7.8 | 5.8 | 5.8 | 5.0 | 5.9 | 5.9 | 5.9 | 5.9 | 0.056 |
| **CHC 12** | 9.5 | 2.9 | 5.2 | 9.4 | 6.5 | 7.5 | 7.5 | 3.4 | 5.5 | 1.9 | 4.5 | 3.4 | 5.5 | 6.9 | 6.9 | 6.9 | 6.9 | 0.022 |
| **CHC 13** | 5.9 | 5.9 | 5.9 | 6.4 | 9.1 | 5.5 | 5.5 | 2.9 | 5.0 | 7.8 | 5.8 | 5.8 | 5.0 | 5.9 | 5.9 | 5.9 | 5.9 | 0.017 |
| **CHC 14** | 17.0 | 15.6 | 2.0 | 7.2 | 7.2 | 4.0 | 13.4 | 1.4 | 2.4 | 5.5 | 6.7 | 2.4 | 0.8 | 0.7 | 3.4 | 6.7 | 3.4 | 0.007 |
| **CHC 15** | 10.0 | 9.2 | 1.2 | 12.0 | 9.2 | 4.1 | 7.5 | 1.4 | 6.3 | 9.0 | 5.4 | 5.4 | 2.9 | 7.7 | 4.0 | 4.0 | 0.8 | 0.022 |
| **Arithmetic mean** | 9.7 | 8.4 | 5.1 | 8.0 | 7.0 | 5.0 | 7.1 | 4.4 | 6.8 | 6.8 | 4.4 | 4.6 | 5.5 | 4.5 | 4.1 | 4.9 | 3.7 | - |
| **Maximum** | 22.0 | 22.0 | 22.0 | 13.9 | 10.3 | 9.4 | 13.4 | 6.8 | 12.8 | 12.8 | 6.7 | 9.4 | 10.2 | 8.7 | 6.9 | 8.6 | 6.9 | - |
| **Minimum** | 2.7 | 2.9 | 0.6 | 0.5 | 1.2 | 1.5 | 3.2 | 1.3 | 1.0 | 1.5 | 1.2 | 0.5 | 0.8 | 0.2 | 0.5 | 1.5 | 0.6 | - |

***Note.*** (1) Weights were derived by combining category-level and factor-level judgments and adjusted to remove bias from unequal category sizes. (2) All values were normalized to a total of 100 percent and are expressed as percentages (one decimal place). (3) A unified green–yellow gradient color scale (midpoint = 5.88%) visualizes relative weight importance, with greener tones indicating higher values.

**Table D.3**
The normalized factor weights (%) assigned by participants in the FES group (with conditional formatting)

| | Factor risk weight | | | | | | | | | | | | | | | | | CR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A1 | A2 | A3 | B1 | B2 | B3 | B4 | B5 | C1 | C2 | C3 | C4 | C5 | D1 | D2 | D3 | D4 | |
| **Group (GM)** | **8.4** | **7.9** | **1.4** | **9.4** | **4.9** | **5.7** | **11.4** | **5.5** | **3.6** | **5.0** | **4.3** | **7.0** | **4.3** | **2.7** | **6.9** | **7.3** | **4.2** | - |
| **FES 01** | 8.4 | 7.7 | 1.0 | 9.7 | 13.4 | 1.5 | 7.8 | 7.8 | 2.1 | 5.1 | 6.3 | 6.3 | 3.9 | 4.7 | 4.7 | 5.6 | 4.0 | 0.022 |
| **FES 02** | 11.7 | 10.3 | 4.5 | 11.8 | 1.3 | 4.0 | 9.0 | 7.3 | 7.6 | 3.3 | 6.1 | 1.1 | 9.9 | 0.5 | 4.7 | 3.8 | 2.9 | 0.019 |
| **FES 03** | 12.4 | 11.4 | 1.5 | 10.0 | 1.1 | 5.8 | 7.3 | 5.8 | 5.4 | 3.0 | 1.0 | 8.7 | 6.7 | 4.6 | 5.5 | 7.3 | 2.5 | 0.022 |
| **FES 04** | 8.0 | 7.4 | 1.0 | 15.1 | 5.1 | 9.4 | 11.6 | 1.7 | 2.4 | 3.1 | 3.9 | 3.9 | 1.3 | 4.6 | 7.6 | 6.4 | 7.6 | 0.022 |
| **FES 05** | 10.9 | 5.6 | 1.2 | 6.4 | 6.4 | 2.7 | 6.9 | 6.9 | 5.7 | 5.7 | 1.2 | 11.2 | 5.7 | 3.7 | 8.6 | 10.1 | 1.2 | 0.006 |
| **FES 07** | 7.8 | 7.1 | 0.9 | 9.1 | 11.2 | 4.9 | 14.6 | 1.7 | 6.9 | 8.5 | 3.8 | 11.1 | 1.3 | 3.1 | 2.6 | 4.1 | 1.4 | 0.017 |
| **FES 08** | 13.0 | 11.9 | 1.6 | 11.4 | 2.2 | 4.0 | 8.7 | 7.2 | 5.9 | 3.1 | 3.1 | 6.8 | 9.2 | 2.9 | 3.5 | 3.5 | 2.1 | 0.022 |
| **FES 09** | 10.5 | 9.7 | 1.3 | 4.4 | 6.1 | 5.9 | 7.6 | 5.7 | 0.9 | 8.0 | 8.0 | 7.3 | 0.9 | 1.0 | 9.0 | 4.6 | 9.0 | 0.053 |
| **FES 10** | 7.4 | 8.0 | 1.0 | 5.1 | 11.6 | 9.4 | 15.1 | 1.7 | 1.3 | 3.1 | 3.9 | 3.9 | 2.4 | 1.2 | 6.3 | 10.4 | 8.2 | 0.017 |
| **FES 11** | 0.9 | 1.9 | 0.9 | 3.8 | 1.3 | 7.0 | 11.3 | 8.6 | 0.7 | 2.1 | 3.9 | 6.3 | 4.8 | 2.9 | 16.6 | 18.1 | 8.7 | 0.009 |
| **Arithmetic mean** | 9.1 | 8.1 | 1.5 | 8.7 | 6.0 | 5.5 | 10.0 | 5.4 | 3.9 | 4.5 | 4.1 | 6.7 | 4.6 | 2.9 | 6.9 | 7.4 | 4.8 | - |
| **Maximum** | 13.0 | 11.9 | 4.5 | 15.1 | 13.4 | 9.4 | 15.1 | 8.6 | 7.6 | 8.5 | 8.0 | 11.2 | 9.9 | 4.7 | 16.6 | 18.1 | 9.0 | - |
| **Minimum** | 0.9 | 1.9 | 0.9 | 3.8 | 1.1 | 1.5 | 6.9 | 1.7 | 0.7 | 2.1 | 1.0 | 1.1 | 0.9 | 0.5 | 2.6 | 3.5 | 1.2 | - |

***Note.*** (1) Weights were derived by combining category-level and factor-level judgments and adjusted to remove bias from unequal category sizes. (2) All values were normalized to a total of 100 percent and are expressed as percentages (one decimal place). (3) A unified green–yellow gradient color scale (midpoint = 5.88%) visualizes relative weight importance, with greener tones indicating higher values.

**Table D.4**
The normalized factor weights (%) assigned by participants in the HBM group (with conditional formatting)

| | Factor risk weight | | | | | | | | | | | | | | | | | CR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A1 | A2 | A3 | B1 | B2 | B3 | B4 | B5 | C1 | C2 | C3 | C4 | C5 | D1 | D2 | D3 | D4 | |
| **Group (GM)** | **11.4** | **9.1** | **6.3** | **2.2** | **10.9** | **7.7** | **6.7** | **6.8** | **1.8** | **4.5** | **5.1** | **5.0** | **1.7** | **2.4** | **6.0** | **7.3** | **5.3** | - |
| **HBM 01** | 19.4 | 19.4 | 2.2 | 7.3 | 10.4 | 6.3 | 6.3 | 3.3 | 2.3 | 1.8 | 1.8 | 1.6 | 0.8 | 2.4 | 4.9 | 4.9 | 4.9 | 0.017 |
| **HBM 02** | 8.8 | 7.0 | 5.5 | 1.2 | 10.3 | 6.4 | 7.9 | 3.5 | 3.5 | 6.4 | 10.3 | 7.9 | 1.2 | 2.5 | 7.3 | 5.6 | 4.6 | 0.056 |
| **HBM 03** | 5.3 | 17.5 | 9.6 | 1.2 | 7.8 | 10.2 | 3.4 | 6.3 | 1.9 | 5.5 | 3.4 | 4.2 | 0.6 | 3.3 | 6.6 | 6.6 | 6.6 | 0.010 |
| **HBM 04** | 10.5 | 4.6 | 12.1 | 1.4 | 12.1 | 4.1 | 7.5 | 9.3 | 1.7 | 3.7 | 3.2 | 5.0 | 1.7 | 2.9 | 6.4 | 8.4 | 5.4 | 0.019 |
| **HBM 05** | 14.3 | 4.3 | 7.9 | 3.3 | 9.2 | 7.3 | 7.3 | 6.4 | 3.3 | 6.1 | 7.6 | 9.9 | 1.1 | 0.7 | 3.3 | 6.1 | 1.9 | 0.017 |
| **HBM 06** | 9.2 | 8.1 | 3.5 | 4.1 | 7.6 | 12.2 | 1.4 | 9.3 | 1.9 | 4.5 | 5.6 | 5.6 | 3.4 | 3.0 | 7.4 | 6.6 | 6.6 | 0.022 |
| **HBM 07** | 11.2 | 8.9 | 7.1 | 1.4 | 7.5 | 12.1 | 4.1 | 9.3 | 1.8 | 3.4 | 4.1 | 5.4 | 0.6 | 2.9 | 6.4 | 8.4 | 5.4 | 0.056 |
| **HBM 08** | 10.9 | 8.7 | 6.9 | 3.3 | 10.4 | 6.3 | 7.3 | 6.3 | 2.5 | 6.0 | 7.5 | 7.5 | 4.6 | 1.5 | 2.8 | 3.3 | 4.4 | 0.056 |
| **HBM 09** | 11.2 | 8.9 | 7.1 | 1.4 | 12.1 | 4.1 | 7.5 | 9.3 | 0.6 | 1.8 | 5.4 | 4.1 | 3.4 | 1.5 | 4.3 | 8.3 | 9.0 | 0.056 |
| **HBM 10** | 14.7 | 8.1 | 4.4 | 1.4 | 12.1 | 4.1 | 9.3 | 7.5 | 0.6 | 5.4 | 4.1 | 3.4 | 1.8 | 1.5 | 9.0 | 8.3 | 4.3 | 0.017 |
| **HBM 11** | 11.0 | 11.0 | 5.5 | 1.2 | 10.3 | 7.9 | 6.4 | 3.5 | 1.8 | 4.2 | 3.4 | 5.5 | 0.6 | 6.5 | 7.7 | 10.2 | 3.4 | 0.017 |
| **HBM 12** | 6.8 | 5.4 | 4.3 | 1.7 | 11.6 | 9.4 | 15.1 | 5.1 | 0.6 | 3.9 | 5.1 | 1.7 | 3.2 | 3.2 | 6.1 | 9.5 | 7.2 | 0.056 |
| **Arithmetic mean** | 11.1 | 9.3 | 6.3 | 2.4 | 10.1 | 7.5 | 7.0 | 6.6 | 1.9 | 4.4 | 5.1 | 5.2 | 1.9 | 2.7 | 6.0 | 7.2 | 5.3 | - |
| **Maximum** | 19.4 | 19.4 | 12.1 | 7.3 | 12.1 | 12.2 | 15.1 | 9.3 | 3.5 | 6.4 | 10.3 | 9.9 | 4.6 | 6.5 | 9.0 | 10.2 | 9.0 | - |
| **Minimum** | 5.3 | 4.3 | 2.2 | 1.2 | 7.5 | 4.1 | 1.4 | 3.3 | 0.6 | 1.8 | 1.8 | 1.6 | 0.6 | 0.7 | 2.8 | 3.3 | 1.9 | - |

***Note.*** (1) Weights were derived by combining category-level and factor-level judgments and adjusted to remove bias from unequal category sizes. (2) All values were normalized to a total of 100 percent and are expressed as percentages (one decimal place). (3) A unified green–yellow gradient color scale (midpoint = 5.88%) visualizes relative weight importance, with greener tones indicating higher values.

### *D.2 Group-consensus and inter-rater reliability*

**Table D.5**
Consensus, distribution, reliability, and rank-order agreement statistics for each group

| Group | Number | Consensus and distribution metrics | | | | Reliability | Rank-order agreement | |
|---|---|---|---|---|---|---|---|---|
| | | $S^*$ | $S$ | $H_\beta$ | $H_\alpha$ | $\alpha$ | $W$ | $p$ |
| FSE | 10 | 0.857 | 0.870 | 0.131 | 2.626 | 0.801 | 0.388 | $< 0.01$ |
| HBM | 12 | 0.914 | 0.922 | 0.077 | 2.648 | 0.928 | 0.598 | $< 0.01$ |
| CHC | 15 | 0.851 | 0.865 | 0.136 | 2.658 | 0.755 | 0.233 | $< 0.01$ |
| BSE | 16 | 0.864 | 0.876 | 0.124 | 2.675 | 0.763 | 0.221 | $< 0.01$ |

***Note.*** $S^*$ is the AHP-specific group consensus indicator, and $S$ is the relative homogeneity index. $H_\beta$ is the Shannon between-rater entropy, reflecting heterogeneity among raters. $H_\alpha$ is the Shannon within-rater entropy, representing the average distribution of attention across criteria. Cronbach's alpha ($\alpha$) represents the internal consistency reliability coefficient. $W$ is Kendall's coefficient of concordance, and $p$-value denotes the significance level of concordance.

### *D.3 Sensitivity analysis of factor weights*

**Table D.6**
Summary of sensitivity analysis across factor weight perturbation scenarios

| Perturbation level | Scenarios | Mandarin's House rank #1 | Ranking changes | Mean Spearman $\rho$ |
|---|---|---|---|---|
| ±10 % | 34 (17 factors × 2 perturbations) | 34/34 (100 %) | 0 | 1.00 |
| ±20 % | 34 (17 factors × 2 perturbations) | 34/34 (100 %) | 0 | 1.00 |
| ±30 % | 34 (17 factors × 2 perturbations) | 34/34 (100 %) | 0 | 1.00 |
| **Overall** | 102 | 102/102 (100 %) | 0 | 1.00 |

The risk indices varied only slightly when the weights were perturbed. The ranges across all scenarios were 38.5–41.8 for Mandarin's House (baseline 40.12), 33.2–35.5 for Lou Kau Mansion (baseline 34.3), and 32.4–34.5 for St. Augustine's Church (baseline 33.5). The Spearman correlations ($\rho = 1.00$) indicate that Mandarin's House consistently ranked as the highest-risk building, confirming the robustness of the prioritization.

## Data availability

Data will be made available on request.

## References


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